ACADEMIC REGULATIONS & SYLLABUS

Faculty of Pharmacy
Master of Pharmacy Programme

(Pharmaceutical Quality Assurance)

(AS PER PCI SYLLABUS)

Ramanbhai Patel College of Pharmacy

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Vision of RPCP

To Become a Premier Pharma Institute by Creating World Class Pharmacists and Researchers.

Mission of RPCP

To Strive for the Excellence in Pharmaceutical Sciences through Quality Education and Research.

PROGRAM OUTCOMES

1. Pharmacy Knowledge: Possess knowledge and comprehension of the core and basic knowledge associated with the profession of pharmacy, including biomedical sciences; pharmaceutical sciences; behavioral, social, and administrative pharmacy sciences; and manufacturing practices.

2. Planning Abilities: Demonstrate effective planning abilities including time management, resource management, delegation skills and organizational skills. Develop and implement plans and organize work to meet deadlines.

3. Problem analysis: Utilize the principles of scientific enquiry, thinking analytically, clearly and critically, while solving problems and making decisions during daily practice. Find, analyze, evaluate and apply information systematically and shall make defensible decisions.

4. Modern tool usage: Learn, select, and apply appropriate methods and procedures, resources, and modern pharmacy-related computing tools with an understanding of the limitations.

5. Leadership skills: Understand and consider the human reaction to change, motivation issues, leadership and team-building when planning changes required for fulfillment of practice, professional and societal responsibilities. Assume participatory roles as responsible citizens or leadership roles when appropriate to facilitate improvement in health and wellbeing.

6. Professional Identity: Understand, analyze and communicate the value of their professional roles in society (e.g. health care professionals, promoters of health, educators, managers, employers, employees).

7. Pharmaceutical Ethics: Honour personal values and apply ethical principles in professional and social contexts. Demonstrate behavior that recognizes cultural and personal variability in values, communication and lifestyles. Use ethical frameworks; apply ethical principles while making decisions and take responsibility for the outcomes associated with the decisions.

8. Communication: Communicate effectively with u the pharmacy community and with society at large, such as, being able to comprehend and write effective reports, make effective presentations and documentation, and give and receive clear instructions.

9. The Pharmacist and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety and legal issues and the consequent responsibilities relevant to the professional pharmacy practice.

10. Environment and sustainability: Understand the impact of the professional pharmacy solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

11. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. Self-assess and use feedback effectively from others to identify learning needs and to satisfy these needs on an ongoing basis.

FACULTY OF PHARMACY

ACADEMIC REGULATIONS
MASTER OF PHARMACY (M. Pharm.) PROGRAMME
Choice Based Credit System (CBCS)

1. Short Title and Commencement

These regulations shall be called as "The Revised Academic Regulations for the postgraduate programmes under the Faculty of Pharmacy". They shall come into effect from the Academic Year 2018-19. The regulations framed are subject to modifications from time to time by the respective regulatory bodies.

2. Minimum Qualification for Admission

2.1 Candidate shall have passed B. Pharm Degree examination of an Indian university established by law in India from an institution approved by Pharmacy Council of India and has scored not less than 55 % of the maximum marks (aggregate of 4 years of B.Pharm.)

2.2 Every student, selected for admission to post graduate pharmacy program in any PCI approved institution should have obtained registration with the State Pharmacy Council or should obtain the same within one month from the date of his/her admission, failing which the admission of the candidate shall be cancelled.

Note: It is mandatory to submit a migration certificate obtained from the respective university where the candidate had passed his/her qualifying degree (B.Pharm.).

3. Duration of the Programme

The course of study for M.Pharm shall extend over a period of four semesters (two academic years). The curricula and syllabi for the program shall be prescribed from time to time by Pharmacy Council of India, New Delhi.

4. Medium of Instruction and Examinations

Medium of instruction and examination shall be in English.

5. Working Days in a Semester

Each semester shall consist of not less than 100 working days. The odd semesters shall be conducted from the month of June/July to November/December and the even semesters shall be conducted from December/January to May/June in every calendar year.

6. Attendance and Progress

A candidate is required to put in at least 80% attendance in individual courses considering theory and practical separately. The candidate shall complete the prescribed course satisfactorily to be eligible to appear for the respective examinations.

7. Programme Credit Structure

As per the philosophy of Credit Based Semester System, certain quantum of academic work viz. theory classes, tutorial hours, practical classes, etc. are measured in terms of credits. On satisfactory completion of the courses, a candidate earns credits. The amount of credit associated with a course is dependent upon the number of hours of instruction per week in that course. Similarly, the credit associated with any of the other academic, co/extra-curricular activities is dependent upon the quantum of work expected to be put in for each of these activities per week.

7.1. Credit assignment

7.1.1. Theory and Laboratory courses

Courses are broadly classified as Theory and Practical. Theory courses consist of lecture (L) and Practical (P) courses consist of hours spent in the laboratory. Credits (C) for a course is dependent on the number of hours of instruction per week in that course, and is obtained by using a multiplier of one (1) for lecture and a multiplier of half (1/2) for practical (laboratory) hours. Thus, for example, a theory course having four lectures per week throughout the semester carries a credit of 4. Similarly, a practical having four laboratory hours per week throughout semester carries a credit of 2.

The contact hours of seminars, assignments and research work shall be treated as that of practical courses for the purpose of calculating credits. i.e., the contact hours shall be multiplied by 1/2. Similarly, the contact hours of journal club, research work presentations and discussions with the supervisor shall be considered as theory course and multiplied by 1.

7.2. Minimum credit requirements

The minimum credit points required for the award of M. Pharm. degree is 95. However based on the credit points earned by the students under the head of co-curricular activities, a student shall earn a maximum of 100 credit points. These credits are divided into Theory courses, Practical, Seminars, Assignments, Research work, Discussions with the supervisor, Journal club and Co-Curricular activities over the duration of four semesters. The credits are distributed semester-wise. Courses generally progress in sequence, building competencies and their positioning indicates certain academic maturity on the part of the learners. Learners are expected to follow the semester-wise schedule of courses given in the syllabus.

8. Academic work

A regular record of attendance both in Theory, Practical, Seminar, Assignment, Journal club, Discussion with the supervisor, Research work presentation and Dissertation shall be maintained by the department / teaching staff of respective courses**.**

9. Examinations/Assessments

The scheme for internal assessment and end semester examinations is given in Annexure II (Table – 1 to 5).

9.1. End Semester Examinations

The End Semester Examinations for each theory and practical course through semesters I to IV shall be conducted by the University for which Examinations shall be conducted by the subject experts at college level and the marks/grades shall be submitted to the university.

9.2. Internal Assessment: Continuous Mode

The marks allocated for Continuous mode of Internal Assessment shall be awarded as per the scheme given below.

Table- 1: Scheme for awarding internal assessment: Continuous mode

Theory	
Criteria	**Maximum Marks**
Attendance (Refer Table – 2)	8
Student – Teacher interaction	2
Total	10
Practical	
Attendance (Refer Table – 2)	10
Based on Practical Records, Regular viva voce, etc.	10
Total	20

Table- 2: Guidelines for the allotment of marks for attendance

Percentage of Attendance	**Theory**	**Practical**
95 – 100	8	10
90 – 94	6	7.5
85 – 89	4	5
80 – 84	2	2.5
Less than 80	0	0

9.2.1. Sessional Exams

Two Sessional exams shall be conducted for each theory / practical course as per the schedule fixed by the college. The scheme of question paper for theory and practical Sessional examinations will be as prescribed by the regulatory body. The average marks of two Sessional exams shall be computed for internal assessment as per the requirements given in Annexure II. Sessional exam shall be conducted for 30 marks for theory and shall be computed for 15 marks. Similarly Sessional exam for practical shall be conducted for 40 marks and shall be computed for 10 marks.

10. Promotion and Award of Grades

A student shall be declared PASS and eligible for getting grade in a course of M.Pharm. Programme, if he/she secures at least 50% marks in that particular course including internal assessment.

11. Carry Forward of Marks

In case a student fails to secure the minimum 50% in any Theory or Practical course as specified in 9 above, then he/she shall reappear for the end semester examination of that course. However his/her marks of the Internal Assessment shall be carried over and he/she shall be entitled for grade obtained by him/her on passing.

12. Improvement of Internal Assessment

A student shall have the opportunity to improve his/her performance only once in the Sessional exam component of the internal assessment. The re-conduct of the Sessional exam shall be completed before the commencement of next end semester theory examinations.

13. Re-examination of End Semester Examinations

Re-examination of end semester examination shall be conducted as per the schedule given in table 3. The exact dates of examinations shall be notified from time to time.

Table-3: Tentative schedule of end semester examinations

Semester	For Regular Candidates	For Failed Candidates
I and III	November / December	May / June
II and IV	May / June	November / December

14. Academic Progression

No student shall be admitted to any examination unless he/she fulfils the norms given in item no. 6 under the heading of attendance and progress. ATKT rules are applicable as follows:

A student shall be eligible to carry forward all the courses of I and II semesters till the III semester examinations. However, he/she shall not be eligible to attend the courses of IV semester until all the courses of I, II and III semesters are successfully completed.

A student shall be eligible to get his/her CGPA upon successful completion of the courses of I to IV semesters within the stipulated time period as per the norms.

Note: Grade AB should be considered as failed and treated as one head for deciding ATKT. Such rules are also applicable for those students who fail to register for examination(s) of any course in any semester.

15. Grading of Performances (Letter Grades and Grade Points Allocations)

Based on the performances, each student shall be awarded a final letter grade at the end of the semester for each course. The letter grades and their corresponding grade points are given in Table – 4.

Table – 4: Letter grades and grade points equivalent to Percentage of marks and performances

Percentage of Marks Obtained	Letter Grade	Grade Point	Performance
90.00 – 100	AA	10	Outstanding
80.00 – 89.99	AB	9	Excellent
70.00 – 79.99	BB	8	Good
60.00 – 69.99	BC	7	Fair
50.00 – 59.99	CC	6	Average
Less than 50	FF	0	Fail
Absent	Ab	0	Fail

A learner who remains absent for any end semester examination shall be assigned a letter grade of Ab. and a corresponding grade point of zero. He/she should reappear for the said evaluation/examination in due course.

16. Semester Grade Point Average (SGPA)

The performance of a student in a semester is indicated by a number called 'Semester Grade Point Average' (SGPA). The SGPA is the weighted average of the grade points obtained in all the courses by the student during the semester. For example, if a student takes five courses(Theory/Practical) in a semester with credits C1, C2, C3, C4 and C5 and the student's grade points in these courses are G1, G2, G3, G4 and G5, respectively, and then students' SGPA is equal to:

$$\text{SGPA}=\frac{C1G1+C2G2+C3G3+C4G4}{C1+C2+C3+C4}$$

The SGPA is calculated to two decimal points. It should be noted that, the SGPA for any semester shall take into consideration the FF and Ab. grade awarded in that semester. For example if a learner has a FF or Ab. grade in course 4, the SGPA shall then be computed as:

$$\text{SGPA}=\frac{C1G1+C2G2+C3G3+C4Zero}{C1+C2+C3+C4}$$

17. Cumulative Grade Point Average (CGPA)

The CGPA is calculated with the SGPA of all the VIII semesters to two decimal points and is indicated in final grade report card/final transcript showing the grades of all VIII semesters and their courses. The CGPA shall reflect the failed status in case of FF grade(s), till the course(s) is/are passed. When the course(s) is/are passed by obtaining a pass grade on subsequent examination(s) the CGPA shall only reflect the new grade and not the fail grades earned earlier. The CGPA is calculated as:

$$\textbf{CGPA}=\frac{\boldsymbol{C1S1+C2S2+C3S3+C4S4}}{\boldsymbol{C1+C2+C3+C4}}$$

where C1, C2, C3,…. is the total number of credits for semester I,II,III,…. and S1,S2, S3,….is the SGPA of semester I,II,III,…. .

No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.

18. Declaration of Class (Table-5)

The class shall be awarded on the basis of CGPA as follows:

First Class with Distinction	= CGPA of. 7.50 to 10.0
First Class	= CGPA of 6.0 to 7.49
Second Class	= CGPA of 5.0 to 5.99
Pass Class	=CGPA of 5.00

19. Project Work

All the students shall undertake a project under the supervision of a teacher in Semester III to IV and submit a report. 4 copies of the project report shall be submitted (typed & bound copy not less than 75 pages).

The internal and external examiner appointed by the University shall evaluate the project at the time of the Practical examinations of other semester(s). The projects shall be evaluated as per the criteria given below.

<table>
<tr><th colspan="5">Evaluation of Dissertation Book</th></tr>
<tr><th rowspan="2">Criteria</th><th colspan="2">Semester-III</th><th colspan="2">Semester-IV</th></tr>
<tr><th colspan="2"></th><th>Internal Evaluation (Marks)</th><th>External Evaluation (Marks)</th></tr>
<tr><td>Objective(s) of the work done</td><td colspan="2">--</td><td>05</td><td>05</td></tr>
<tr><td>Methodology Adopted</td><td colspan="2">--</td><td>25</td><td>25</td></tr>
<tr><td>Results and Discussions</td><td colspan="2">--</td><td>15</td><td>15</td></tr>
<tr><td>Conclusions and Outcome</td><td colspan="2">--</td><td>05</td><td>05</td></tr>
<tr><td>Total</td><td colspan="2">--</td><td>50</td><td>50</td></tr>
<tr><td>Final Total</td><td colspan="2"></td><td colspan="2">100</td></tr>
<tr><th colspan="5">Evaluation of Presentation</th></tr>
<tr><th></th><th colspan="2">Semester-III</th><th colspan="2">Semester-IV</th></tr>
<tr><th>Criteria</th><th>Internal Evaluation (Marks)</th><th>External Evaluation (Marks)</th><th>Internal Evaluation (Marks)</th><th>External Evaluation (Marks)</th></tr>
<tr><td>Presentation of work</td><td>75</td><td>100</td><td>75</td><td>75</td></tr>
<tr><td>Communication skills</td><td>25</td><td>50</td><td>25</td><td>25</td></tr>
<tr><td>Question and answer skills</td><td>50</td><td>50</td><td>50</td><td>50</td></tr>
<tr><td>Total</td><td>150</td><td>200</td><td>150</td><td>150</td></tr>
<tr><td>Final Total</td><td colspan="2">350</td><td colspan="2">300</td></tr>
</table>

20. Award of Ranks

Ranks and Medals shall be awarded on the basis of final CGPA. However, candidates who fail in one or more courses during the M. Pharm. program shall not be eligible for award of ranks.

Moreover, the candidates should have completed the M. Pharm program in minimum prescribed number of years, (two years) for the award of Ranks.

21. Award of degree

Candidates who fulfil the requirements mentioned above shall be eligible for award of degree during the ensuing convocation.

22. Duration for completion of the program of study

The duration for the completion of the program shall be fixed as double the actual duration of the program and the students have to pass within the said period, otherwise they have to get fresh Registration.

23. Extra Credit:

An extra credit is to be offered to a student for achievements in co-curricular and extra–curricular activities. This credit shall not be counted while considering the minimum credits for completing the program. The activities and appropriate weight (points) to be allocated to award an extra credit are broadly classified as per the tables below:

Extracurricular Activities (25 Marks maximum out of 100 marks) includes

Sr. No.	Name of the activity	Maximum Marks
1	Participation in any Sport	10
2	Participation in any Cultural event	10
3	Community and Extension activities	10
4	Any achievement includes award, prize or special recognition for event	05

Co-Curricular Activities (75 marks maximum out of 100 marks) include

Sr. No.	Name of the activity	Maximum Marks
1	Completion of certificate SWAYAM course (minimum 4 weeks)	20 (10 marks/course, given for maximum two course)
2	Participation in Conference/Seminar/training program/Teach fest etc. (without Paper) includes	10
	a. State/University/Zonal level: 03	
	b. National : 05	
	c. International conference : 07	
3	Presentation in Conference/Seminar/training program/Teach fest etc. includes	15
	a. Oral Presentation: 10	

	b. Poster Presentation: 05	
4	Publication in Peer-reviewed Journal (indexed in Web of Science and Scopus):	20
	a. First paper publication: 10	
	b. Second paper publication:10	
5	Participation in other technical events like model preparation, Rangoli, Painting, Pharma receipt, dare to sell etc.: (05)	05
6	Any achievement includes award, prize or special recognition for any technical events mentioned above: (05)	05

Activities are broadly classified into Extra-curricular and Co-curricular. All suggested activities are counted as total sum of 100 marks.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
MASTER OF PHARMACY (PHARMACEUTICAL QUALITY ASSURANCE) PROGRAMME
Schemes for internal assessments and end semester examinations

SEMESTER-I
SCHEME OF TEACHING

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
PHCCC001	Modern Pharmaceutical Analytical Techniques	4	4	4
PHPQA001	Quality Management System	4	4	4
PHPQA002	Quality Control and Quality Assurance	4	4	4
PHCCC007	Hazards, Safety and Risk Management & Intellectual Property Rights	4	4	4
PHCCC002	Drug Development and Pharmaceutical Statistics	2	2	2
PHPQA003	Quality Assurance Practical-I	12	6	12
PHPQA004	Seminar/Assignment-I	2	1	2
---	DHSS Elective-I*	2	2	2
---	University Elective-I**	2	2	2
	Total	**36**	**29**	**36**

SCHEME OF EVALUATION

Course Code	Course	Internal Assessment				End Semester Exams		Total Marks
		Continuous Mode	Sessional Exams		Total	Marks	Duration	
			Marks	Duration				
PHCCC001	Modern Pharmaceutical Analytical Techniques	10	15	1Hr	**25**	**75**	3Hrs	**100**
PHPQA001	Quality Management System	10	15	1Hr	**25**	**75**	3Hrs	**100**
PHPQA002	Quality Control and Quality Assurance	10	15	1Hr	**25**	**75**	3Hrs	**100**
PHCCC007	Hazards, Safety and Risk Management & Intellectual Property Rights	10	15	1Hr	**25**	**75**	3Hrs	**100**
PHCCC002	Drug Development and Pharmaceutical Statistics	10	15	1Hr	**25**	**75**	3Hrs	**100**
PHPQA003	Quality Assurance Practical-I	20	30	6Hrs	**50**	**100**	6Hrs	**150**
PHPQA004	Seminar/Assignment-I	-	-	-	**100**	**-**	-	**100**
---	DHSS elective-I*	-	-	-	**30**	**70**	-	**100**
---	University Elective-I**	-	-	-	**30**	**70**	-	**100**
Total								**950**

*DHSS elective courses: SCHEME OF TEACHING

Semester	Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
I	HS704 A/B/C/D/E/F/G/H	Academic Speaking	02	02	02

*DHSS elective courses: SCHEME OF EVALUATION

Course Code	Course Name	Evaluation Scheme				
		Theory		Practical		Total
		Internal	External	Internal	External	
HS704 A/B/C/D/E/F/G/H	Academic Speaking	-	-	30	70	100

**University elective courses: SCHEME OF TEACHING - Semester-I

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
MA771.01	Reliability and Risk Analysis-I	2	2	2
EE781.01	Optimization Techniques	2	2	2
CE772.01	Research Methodology	2	2	2
CA730	Internet and Web Designing	2	2	2
PT795.01	Heath and Physical activity	2	2	2
NR755	First aid and Life support	2	2	2
RD701.01	Introduction to analytical Techniques	2	2	2
RD702.01	Introduction to Nanoscience and Technology	2	2	2
MB650	Creative Leadership	2	2	2
PH891	Community Pharmacy ownership	2	2	2
ME781.01	Occupational Health and Safety	2	2	2
PD261	Astrophysics, Space and Cosmos-1(ASC-1)	2	2	2

**University elective courses: SCHEME OF EVALUATION- Semester-I

Course Code	Course Name	Evaluation Scheme				
		Theory		Practical		Total
		Internal	External	Internal	External	
MA771.01	Reliability and Risk Analysis-I	-	-	30	70	100
EE781.01	Optimization Techniques	-	-	30	70	100
CE772.01	Research Methodology	-	-	30	70	100
CA730	Internet and Web Designing	-	-	30	70	100
PT795.01	Heath and Physical activity	-	-	30	70	100
NR755	First aid and Life support	-	-	30	70	100
RD701.01	Introduction to analytical Techniques	-	-	30	70	100
RD702.01	Introduction to Nanoscience and Technology	-	-	30	70	100
MB650	Creative Leadership	-	-	30	70	100
PH891	Community Pharmacy ownership	-	-	30	70	100
ME781.01	Occupational Health and Safety	-	-	30	70	100
PD261	Astrophysics, Space and Cosmos-1(ASC-1)	-	-	30	70	100

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
MASTER OF PHARMACY (PHARMACEUTICAL QUALITY ASSURANCE) PROGRAMME
Schemes for internal assessments and end semester examinations

SEMESTER-II
SCHEME OF TEACHING

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
PHPQA005	Product Development and Technology Transfer	4	4	4
PHCCC008	Pharmaceutical Validation and Documentation	4	4	4
PHPQA006	Audit and Regulatory Compliance	4	4	4
PHPQA007	Pharmaceutical Manufacturing Technology	4	4	4
PHCCC004	Research Methodology	2	2	2
PHPQA008	Quality Assurance Practical-II	12	6	12
PHPQA009	Seminar/Assignment-II	2	1	2
---	DHSS elective-II*	2	2	2
---	University Elective-II**	2	2	2
	Total	**36**	**29**	**36**

SCHEME OF EVALUATION

Course Code	Course	Internal Assessment				End Semester Exams		Total Marks
		Continuous Mode	Sessional Exams		Total	Marks	Duration	
			Marks	Duration				
PHPQA005	Product Development and Technology Transfer	10	15	1Hr	25	75	3Hrs	100
PHCCC008	Pharmaceutical Validation and Documentation	10	15	1Hr	25	75	3Hrs	100
PHPQA006	Audit and Regulatory Compliance	10	15	1Hr	25	75	3Hrs	100
PHPQA007	Pharmaceutical Manufacturing Technology	10	15	1Hr	25	75	3Hrs	100
PHCCC004	Research Methodology	10	15	1Hr	25	75	3Hrs	100
PHPQA008	Quality Assurance Practical-II	20	30	6Hrs	50	100	6Hrs	150
PHPQA009	Seminar/Assignment-II	-	-	-	100	-	-	100
---	DHSS elective*	-	-	-	30	70	-	100
---	University Elective**	-	-	-	30	70	-	100
Total								**950**

*DHSS elective courses: SCHEME OF TEACHING

Semester	Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
II	HS705 (B)	Academic Writing	02	02	02

*DHSS elective courses: SCHEME OF EVALUATION

Course Code	Course Name	Evaluation Scheme				
		Theory		Practical		Total
		Internal	External	Internal	External	
HS705 (B)	Academic Writing	-	-	30	70	100

**University elective courses: SCHEME OF TEACHING - Semester-II

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
EE782.01	Energy Auditing and Management	2	2	2
CE771.01	Project Management	2	2	2
IT771.01	Cyber Security & Laws	2	2	2
CA842	Mobile Application Development	2	2	2
PT796.01	Fitness and Nutrition	2	2	2
NR 752.01	Epidemiology & Community Heath	2	2	2
OC733.01	Introduction to Polymer Science	2	2	2
MB651	Software based Statistical Analysis	2	2	2
MA772.01	Design of Experiments	2	2	2
PH892	Intellectual Property Rights	2	2	2
PD262	Astrophysics, Space and Cosmos-2 (ASC-2)	2	2	2

**University elective courses: SCHEME OF EVALUATION- Semester-II

Course Code	Course Name	Evaluation Scheme				
		Theory		Practical		Total
		Internal	External	Internal	External	
EE782.01	Energy Auditing and Management	-	-	30	70	100
CE771.01	Project Management	-	-	30	70	100
IT771.01	Cyber Security & Laws	-	-	30	70	100
CA842	Mobile Application Development	-	-	30	70	100
PT796.01	Fitness and Nutrition	-	-	30	70	100
NR 752.01	Epidemiology & Community Heath	-	-	30	70	100
OC733.01	Introduction to Polymer Science	-	-	30	70	100
MB651	Software based Statistical Analysis	-	-	30	70	100
MA772.01	Design of Experiments	-	-	30	70	100
PH892	Intellectual Property Rights	-	-	30	70	100
PD262	Astrophysics, Space and Cosmos-2 (ASC-2)	-	-	30	70	100

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

MASTER OF PHARMACY (PHARMACEUTICAL QUALITY ASSURANCE) PROGRAMME

Schemes for internal assessments and end semester examinations

SEMESTER-III

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
PHPQA010	Dissertation Part-I	32	16	32
PHCCC005	Journal Club-I	02	02	02
	Total	**34**	**18**	**34**

SCHEME OF EVALUATION

Course Code	Course	Internal Assessment				End Semester Exams		Total Marks
		Continuous Mode	Sessional Exams		Total	Marks	Duration	
			Marks	Duration				
PHPQA010	Dissertation Part-I	-	-	-	150	200	-	350
PHCCC005	Journal Club-I	-	-	-	25	-	-	25
Total								375

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

MASTER OF PHARMACY (PHARMACEUTICAL QUALITY ASSURANCE) PROGRAMME

Schemes for internal assessments and end semester examinations

SEMESTER -IV

Course Code	Course Name	Credit Hours	Credit Points	Hrs/Week
PHPQA011	Dissertation Part-II	32	16	32
PHCCC006	Journal Club-II	2	2	2
	Total	**34**	**18**	**34**

SCHEME OF EVALUATION

Course Code	Course	Internal Assessment				End Semester Exams		Total Marks
		Continuous Mode	Sessional Exams		Total	Marks	Duration	
			Marks	Duration				
PHPQA011	Dissertation Part-II	-	-	-	200	200	-	400
PHCCC006	Journal Club-II	-	-	-	25	-	-	25
Total								425

FACULTY OF PHARMACY
Master of Pharmacy Programme

Syllabi
Semester I

Charotar University of Science and Technology

FACULTY OF PHARMACY
Bachelor of Pharmacy Programme
PHCCC001: Modern Pharmaceutical Analytical Techniques (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	**(a)** UV-Visible spectroscopy	12
	(b) IR spectroscopy	
	(c) Spectroflourimetry	
	(d) Flame emission spectroscopy and Atomic absorption spectroscopy	
2	NMR Spectroscopy	12
3	**(a)** Mass Spectroscopy	12
	(b) X ray Crystallography	
4	Chromatography	12
5	**(a)** Electrophoresis	12
	(b) Immunological Assays	
	(c) Estimation of Drugs from Biological Fluids	

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**(a) UV-Visible spectroscopy**	**12**	**20%**
	• UV-Visible spectroscopy: Introduction, Theory, Laws, Instrumentation associated with UV-Visible spectroscopy, Choice of solvents and solvent effect and Applications of UV Visible spectroscopy.		
	(b) IR spectroscopy		
	• Theory, Modes of Molecular vibrations, Sample handling, Instrumentation of Dispersive and Fourier - Transform IR Spectrometer, Factors affecting vibrational frequencies and		

	Applications of IR spectroscopy		
	(c) Spectroflourimetry		
	• Theory of Fluorescence, Factors affecting fluorescence, Quenchers, Instrumentation and Applications of fluorescence spectrophotometer.		
	(d) Flame emission spectroscopy and Atomic absorption spectroscopy		
	• Principle, Instrumentation, Interferences and Applications.		
2.	**NMR Spectroscopy**	**12**	**20%**
	• Quantum numbers and their role in NMR, Principle, Instrumentation, Solvent requirement in NMR, Relaxation process, NMR signals in various compounds, Chemical shift, Factors influencing chemical shift, SpinSpin coupling, Coupling constant, Nuclear magnetic double resonance, Brief outline of principles of FT-NMR and 13C NMR. Applications of NMR spectroscopy.		
3.	**(a) Mass Spectroscopy**	**12**	**20%**
	• Mass Spectrometry: Principle, Theory, Instrumentation of Mass Spectrometry, Different types of ionization like electron impact, chemical, field, FAB and MALDI, APCI, ESI, APPI Analysers of Quadrupole and Time of Flight, Mass fragmentation and its rules, Meta stable ions, Isotopic peaks and Applications of Mass spectrometry.		
	(b) X ray Crystallography		
	• Production of X rays, Different X ray diffraction methods, Bragg's law, Rotating crystal technique, X ray powder technique, Types of crystals and applications of X ray diffraction.		
4.	**Chromatography**	**12**	**20%**
	• Principle, apparatus, instrumentation, chromatographic		

	parameters, factors affecting resolution and applications of the following: a) Paper chromatography b) Thin Layer chromatography c) Ion exchange chromatography d) Column chromatography e) Gas chromatography f) High Performance Liquid chromatography g) Affinity chromatography		
5.	**(a) Electrophoresis**	**12**	**20%**
	• Principle, Instrumentation, Working conditions, factors affecting separation and applications of the following: a) Paper electrophoresis b) Gel electrophoresis c) Capillary electrophoresis d) Zone electrophoresis e) Moving boundary electrophoresis f) Iso electric focusing		
	(b) Immunological Assays		
	• RIA (Radio 13mmune assay), ELISA, Bioluminescence assays.		
	(c) Estimation of Drugs from Biological Fluids		
	• Importance, Special Problems with Biological Fluids, Need for and background of analysis of drugs in biological fluids, Problems encountered in testing biological specimens, Extraction procedures for drugs and metabolites from biological samples LLE, SPE, Precipitation etc. Factors affecting extraction of drugs. Case studies of drug analysis from bio fluids		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	describe theory and principle of various spectroscopic and chromatographic separation techniques.
CO2	describe instrumentation and application of various spectroscopic and chromatographic separation techniques with justification.
CO3	summarize approaches to be adopted for quantitative & qualitative analysis of drugs in single and combine dosage forms.
CO4	describe the approaches for analysing the drugs from biological fluids and medias.
CO5	interpret the spectra and propose the structure of organic compounds.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	-	-	-	3
CO2	3	3	3	-	3	-	-	3	-	3	3
CO3	3	3	3	3	-	3	-	3	-	3	3
CO4	3	3	3	3	-	3	-	3	3	3	3
CO5	3	-	3	3	-	-	-	3	-	-	3

❖ **References:**

1. Spectrometric Identification of Organic compounds - Robert M Silverstein, Sixth edition, John Wiley & Sons, 2004.
2. Principles of Instrumental Analysis - Doglas A Skoog, F. James Holler, Timothy A. Nieman, 5th edition, Eastern press, Bangalore, 1998
3. Instrumental methods of analysis – Willards, 7th edition, CBS publishers.
4. Practical Pharmaceutical Chemistry – Beckett and Stenlake, Vol II, 4th edition, CBS Publishers, New Delhi, 1997.
5. Organic Spectroscopy - William Kemp, 3rd edition, ELBS, 1991.
6. Quantitative Analysis of Drugs in Pharmaceutical formulation - P D Sethi, 3rd Edition, CBS Publishers, New Delhi, 1997.
7. Pharmaceutical Analysis- Modern methods – Part B - J W Munson, Volume 11, Marcel Dekker Series. 8. The Analysis of Drugs in Biological Fluids, Joseph Chamberlain, CRC Press

FACULTY OF PHARMACY

Master of Pharmacy Programme
PHPQA001: Quality Management System (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to quality Quality as a strategic design Customer focus Cost of quality	12
2	Pharmaceutical quality Management	12
3	Six sigma inspection model Quality systems	12
4	Drug stability Quality risk management	12
5	Statistical process control Regulatory Compliance through Quality Management and development of Quality Culture Benchmarking	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Introduction to quality, quality as a strategic design, customer focus, cost of quality**	**12**	**20%**
	Introduction to Quality: Evolution of Quality, Definition of Quality, Dimensions of Quality Quality as a Strategic Decision: Meaning of strategy and strategic quality management, mission and vision statements, quality policy, Quality objectives, strategic planning and implementation, McKinsey 7s model, Competitive analysis, Management commitment to quality Customer Focus: Meaning of customer and customer focus, Classification of customers, Customer focus, Customer perception of quality, Factors affecting customer perception, Customer requirements, Meeting customer needs and expectations, Customer satisfaction and Customer delight, Handling customer complaints, Understanding customer behavior, concept of internal and external customers. Case studies.		

	Cost of Quality: Cost of quality, Categories of cost of Quality, Models of cost of quality, Optimising costs, Preventing cost of quality.		
2.	**Pharmaceutical quality Management**	**12**	**20%**
	Basics of Quality Management, Total Quality Management (TQM), Principles of Six sigma, ISO 9001:2008, 9001:2015, ISO 14001:2004, Pharmaceutical Quality Management – ICH Q10, Knowledge management, Quality Metrics, Operational Excellence and Quality Management Review. OSHAS guidelines, NABL certification and accreditation, CFR-21 part 11, WHO-GMP requirements.		
3.	**Six sigma inspection model, quality systems**	**12**	**20%**
	Six System Inspection model: Quality Management system, Production system, Facility and Equipment system, Laboratory control system, Materials system, Packaging and labeling system. Concept of self-inspection. Quality systems: Change Management/ Change control. Deviations, Out of Specifications (OOS), Out of Trend (OOT), Complaints - evaluation and handling, Investigation and determination of root cause, Corrective & Preventive Actions (CAPA), Returns and Recalls, Vendor Qualification, Annual Product Reviews, Batch Review and Batch Release. Concept of IPQC, area clearance/ Line clearance.		
4.	**Drug stability, quality risk management**	**12**	**20%**
	Drug Stability: ICH guidelines for stability testing of drug substances and drug products. Study of ICH Q8, Quality by Design and Process development report Quality risk management: Introduction, risk assessment, risk control, risk review, risk management tools, HACCP, risk ranking and filtering according to ICH Q9 guidelines.		
5.	**Stastical process control, Regulatory Compliance through Quality Management and development of Quality Culture Benchmarking**	**12**	**20%**
	Statistical Process control (SPC): Definition and Importance of SPC, Quality measurement in manufacturing, Statistical control charts - concepts and general aspects, Advantages of statistical control, Process capability, Estimating Inherent or potential capability from a control chart analysis, Measuring process control and quality improvement, Pursuit of decreased process variability.		

	Regulatory Compliance through Quality Management and development of Quality Culture Benchmarking: Definition of benchmarking, Reasons for benchmarking, Types of Benchmarking, Benchmarking process, Advantages of benchmarking, Limitations of benchmarking.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	summarize importance of Quality, tools to improve the quality and analyze the issues in Quality of Pharmaceutical Products.
CO2	narrate the concept of Total Quality Management, Quality systems Approach with reference to pharmaceutical manufacturing.
CO3	describe the importance and application of Quality Risk Management concept in pharmaceutical manufacturing processes.
CO4	describe and compare ICH guidelines for stability testing of drug substances and drug products with reference to similar guidelines issued by other competent authorities.
CO5	describe statistical approaches and benchmarking for quality of pharmaceuticals.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO 11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3
CO5	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

- ❖ **Text book:**
    1. Implementing Juran's Road Map for Quality Leadership: Benchmarks and Results, By Al Endres, Wiley, 2000 Anatomy and Physiology in Health and Illness by Kathleen J.W. Wilson, Churchill, Livingstone, New York
    2. Understanding, Managing and Implementing Quality: Frameworks, Techniques and Cases, By Jiju Antony; David Preece, Routledge, 2002
    3. Organizing for High Performance: Employee Involvement, TQM, Reengineering, and Knowledge Management in the Fortune 1000: The CEO Report By Edward E. Lawler; Susan Albers Mohrman; George Benson, Jossey-Bass, 2001

4. Corporate Culture and the Quality Organization By James W. Fairfield-Sonn, Quorum Books, 2001
5. The Quality Management Sourcebook: An International Guide to Materials and Resources By Christine Avery; Diane Zabel, Routledge, 1997
6. The Quality Toolbox, Second Edition, Nancy R. Tague, ASQ Publications
7. Juran's Quality Handbook, Sixth Edition, Joseph M. Juran and Joseph A. De Feo, ASQ Publications
8. Root Cause Analysis, The Core of Problem Solving and Corrective Action, Duke Okes, 2009, ASQ Publication

FACULTY OF PHARMACY
Master of Pharmacy Programme
PHPQA002: Quality Control & Quality Assurance (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction	12
2	cGMP guideline	12
	USFDA	
	Pharmaceutical Inspection Convention	
	WHO	
	EMEA	
3	Analysis of raw materials, finished products, packaging materials	12
	In-process quality control	
	Developing & Purchase specifications	
	Maintenance of stores for raw materials	
4	Quality Control Laboratory and its activities	12
	Calibration of analytical instruments	
5	Manufacturing operations & contols	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Introduction**	**12**	**20%**
	Concept and evolution and scopes of Quality Control and Quality Assurance, Good Laboratory Practice, GMP, Overview of ICH Guidelines - QSEM, with special emphasis on Q-series guidelines. Good Laboratory Practices: Scope of GLP, Definitions, Quality assurance unit, protocol for conduct of non-clinical testing, control on animal house, report preparation and documentation. CPCSEA guidelines.		
2.	**cGMP guidelines according to schedule M, USFDA (inclusive of CDER and CBER) Pharmaceutical Inspection Convention (PIC), WHO and EMEA covering**	**12**	**20%**
	Organization and personnel responsibilities, training, hygiene and personal records, drug industry location, design, construction and plant lay out, maintenance, sanitation, environmental control, utilities and maintenance of sterile areas, control of contamination		

	and Good Warehousing Practice.		
3.	**Analysis of raw materials, finished products, packaging materials, in process quality control (IPQC), Developing specification (ICH Q6 and Q3), purchase specifications and maintenance of stores for raw materials**	**12**	**20%**
	In process quality control and finished products quality control for following dosage forms in Pharma industry according to Indian, US and British pharmacopoeias: tablets, capsules, ointments, suppositories, creams, parenterals, ophthalmic and surgical products (How to refer pharmacopoeias).		
4.	**Quality Control Laboratory and its activities, Calibration of analytical instruments**	**12**	**20%**
	Quality Control Laboratory and its activities: Lab layout, Requirements of Facilities, Equipments, Personnel, Activities such as Calibration of QC equipments, Testing Procedures, Specification. Calibration of Analytical Instruments: Importance of calibration, frequency of calibration, difference between validation and calibration, calibration of various analytical instruments like pH meter, conductometer, potentiometer, HPLC, HPTLC, GC, LC-MS, UV-visible, Spectroflourimeter, dissolution apparatus, disintegration test apparatus, friability tester, digital weight balance, IR spectrometer, Melting point apparatus, DSC, TGA		
5.	**Manufacturing operations & Controls**	**12**	**20%**
	Sanitation of manufacturing premises, mix-ups and cross contamination, processing of intermediates and bulk products, packaging operations, IPQC, release of finished product, process deviations, charge-in of components, time limitations on production, drug product inspection, expiry date calculation, calculation of yields, production record review, change control, sterile products, aseptic process control, packaging, reprocessing, salvaging, handling of waste and scrap disposal.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	summarize the roles and responsibilities of QC and QA department
CO2	describe the cGMP concept and would be able to compare various cGMP guidelines used in pharmaceutical industry
CO3	suggest the IPQC and FPQC tests in pharmaceutical industry
CO4	narrate the concept and activities of quality control laboratory

CO5	narrate various manufacturing operations along with their controls

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3
CO5	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ **Textbook:**

1. Quality Assurance Guide by organization of Pharmaceutical Procedures of India, 3 rd revised edition, Volume I & II, Mumbai, 1996. Anatomy and Physiology in Health and Illness by Kathleen J.W. Wilson, Churchill, Livingstone, New York
2. Good Laboratory Practice Regulations, 2nd Edition, Sandy Weinberg Vol. 69, Marcel Dekker Series, 1995.
3. T Quality Assurance of Pharmaceuticals- A compendium of Guide lines and Related materials Vol I & II, 2nd edition, WHO Publications, 1999. Principles of Anatomy and Physiology by Tortora Grabowski. Palmetto, GA, U.S.A.
4. How to Practice GMP's – P P Sharma, Vandana Publications, Agra, 1991. 126
5. The International Pharmacopoeia – vol I, II, III, IV & V - General Methods of Analysis and Quality specification for Pharmaceutical Substances, Excipients and Dosage forms, 3 rd edition, WHO, Geneva, 2005.
6. Good laboratory Practice Regulations – Allen F. Hirsch, Volume 38, Marcel Dekker Series, 1989.
7. ICH guidelines.
8. ISO 9000 and total quality management.
9. The drugs and cosmetics act 1940 – Deshpande, Nilesh Gandhi, 4 th edition, Susmit Publishers, 2006.
10. QA Manual – D.H. Shah, 1 st edition, Business Horizons, 2000.
11. Good Manufacturing Practices for Pharmaceuticals a plan for total quality control – Sidney H. Willig, Vol. 52, 3 rd edition, Marcel Dekker Series.
12. Steinborn L. GMP/ISO Quality Audit Manual for Healthcare Manufacturers and Their Suppliers, Sixth Edition, (Volume 1 - With Checklists and Software Package). Taylor & Francis; 2003.

13. Sarker DK. Quality Systems and Controls for Pharmaceuticals. John Wiley & Sons; 2008.
14. Packaging of Pharmaceuticals.
15. Schedule M and Schedule.

FACULTY OF PHARMACY
Master of Pharmacy Programme
PHCCC007: Hazard, Safety and Risk Management & Intellectual Property Rights (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Multidisciplinary nature of environmental studies	12
2	Air and Chemical based hazards	12
3	Fire Explosion hazard and hazard risk management	12
4	General Principles of Intellectual Property	12
5	Patent and Trademark	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Multidisciplinary nature of environmental studies**	**12**	**20%**
	• Natural Resources, Renewable and non-renewable resources, Natural resources and associated problems, a) Forest resources; b) Water resources; c) Mineral resources; d) Energy resources; e) Land resources Ecosystems: Concept of an ecosystem and Structure and function of an ecosystem. • Environmental hazards: Hazards based on Air, Water, Soil and Radioisotopes.		
2.	**Air and Chemical based hazards**	**12**	**20%**
	• Sources, Types of Hazards, Air circulation maintenance industry for sterile area and non-sterile area, Preliminary Hazard Analysis (PHA) Fire protection system: Fire prevention, types of fire extinguishers and critical Hazard management system. • Sources of chemical hazards, Hazards of Organic synthesis, sulphonating hazard, Organic solvent hazard, Control		

	measures for chemical hazards, Management of combustible gases, Toxic gases and Oxygen displacing gases management, Regulations for chemical hazard, Management of over-Exposure to chemicals and TLV concept.		
3.	**Fire Explosion hazard and hazard risk management**	**12**	**20%**
	• Introduction, Safety and hazards regulations, Fire protection system: Fire prevention, types of fire extinguishers and critical Hazard management system mechanical and chemical explosion, multiphase reactions, transport effects and global rates. Preventive and protective management from fires and explosion electricity passivation, ventilation, and sprinkling, proofing, relief systems -relief valves, flares, scrubbers. • Industrial processes and hazards potential, mechanical electrical, thermal and process hazards. Self-protective measures against workplace hazards. Critical training for risk management, Process of hazard management, ICH guidelines on risk assessment and Risk management methods and Tools Factory act and rules, fundamentals of accident prevention, elements of safety programme and safety management, Physicochemical measurements of effluents, BOD, COD, Determination of some contaminants, Effluent treatment procedure, Role of emergency services.		
4.	**General Principles of Intellectual Property**	**12**	**20%**
	• Concepts of Intellectual Property (IP), Intellectual Property Protection (IPP), Intellectual Property Rights (IPR); Economic importance, mechanism for protection of Intellectual Property –patents, Copyright, Trademark; Factors affecting choice of IP protection; Penalties for violation; Role of IP in pharmaceutical industry; Global ramification and financial implications.		

5.	**Patent and Trademark**	**12**	**20%**
	• Definition, Need for patenting, Conditions to be satisfied by an invention to be patentable, Types of patent applications- provisional and Complete Application, PCT and convention patent applications; Filing a patent applications; patent application forms and guidelines • Patent of Addition; International patenting requirement procedures; Rights and responsibilities of a patentee; Practical aspects regarding maintaining of a Patent file; Opposition of Paten (Pre Grant and Post Grant Opposition); Patent infringement meaning and scope. • IP and ethics-positive and negative aspects of IPP; Societal responsibility, avoiding unethical practices, Brief introduction to Trademark and its protection		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	describe the environment and related issues and suggest probable mitigation plans.
CO2	enumerate various hazards associated with the pharmaceutical industries, their management and prevention techniques.
CO3	exhibit comprehensive knowledge on the safety issues associated with the working area and suggest the practices to provide safe working environment.
CO4	describe the concept of intellectual properties as per the laws, components of patent applications and requirements to obtain the patent.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	-	3	3	3
CO2	3	3	3	3	-	-	-	-	-	-	3

CO3	3	3	-	3	-	-	3	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ **References:**

1. Environmental Science,By Y.K. Sing, New Age International Pvt, Publishers, Bangalore
2. Quantitative Risk Assessment in Chemical Process Industries‖ American Institute of Chemical Industries, Centre for Chemical Process safety.
3. Bharucha Erach, The Biodiversity of India, Mapin Pu blishing Pvt. Ltd., Ahmedabad – 380 013, India,
4. Hazardous Chemicals: Safety Management and Global Regulations, T.S.S. Dikshith, CRC press
5. The enforcement of Intellectual Property Right: a case Book,by Louis Tc. Harms, WIPO Publishing Hoouse, Geneva.
6. Intellectual Property Right, By M.M.S. Khatri, Atlantic Publication.
7. Biodiversity and Biotechnology: Intellectual Property Right, By Kumar-Arvind and Das Govind, Narosa Publication.
8. Writing Chemistry Patents and Intellectual Property Right: A Practical Guide, ByFrancis J. Wallar, Willi Pubishers.
9. Intellectual Property Right and Drug Regulatory Affairs, By Ruchi Tiwari, Nirali Prakashan. Pune. India.
10. IPR Handbook for Pharma Students and Researchers, Parikshit Bansal, Pharma Book Syndicate, Hyderabad, India.
11. Patents, N. R. Subbaram, Pharma Book Syndicate, Hyderabad.
12. Intellectual Property Right by Nikolaus Thumm, Springer-Verlag Publications, Germany.
13. The Enforcement of Intellectual Property Rights: A Case Book by Louis Tc. Harms, WIPO Publishing House, Geneva.
14. Intellectual Property: From Creation to Commercialization - A Practical Guide for Innovators & Researchers by John P., MC Manus, Oak Tree Press, Ireland
15. Intellectual Property Right basic Concept, by M. M. S. Khatri, Atlantic Publisher and Distributors Pvt. Ltd., New Delhi.

16. Epstein on Intellectual Property: 5th Edition by Michael M. Epstein, Wolters Kulwer India Pvt. Ltd. Gurgaon, India.
17. Intellectual Property Right and Human Development in India by Shabana Talwar, First Edition, Serials Publications, New Delhi, India.

FACULTY OF PHARMACY
Master of Pharmacy Programme
PHCCC002: Drug Development and Pharmaceutical Statistics (Theory)

Total hours: 30 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	**(a)** Overview of Drug Development Process	10
	(b) FDA Applications	
	(c) Pre-Clinical Development	
2	**(a)** Overview of Clinical Development	10
	(b) Regulatory aspects related to clinical trials	
3	Biostatistics	10

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**(a) Overview of Drug Development Process**	**10**	**33.3%**
	• Drug discovery cycle, Lead identification, Timeline for drug development, New Chemical Entities.		
	(b) FDA Applications		
	• IND, NDA, ANDA.		
	(c) Pre-Clinical Development		
	• Physicochemical properties of NCE, CMC development, pre-clinical testing – toxicity, pharmacokinetics and metabolism of NCE, In vitro methods for pre-clinical studies.		
2.	**(a) Overview of Clinical Development**	**10**	**33.3%**
	• Introduction to clinical trials, Clinical trial phases, Clinical study designs, Ethical considerations, Inclusion and Exclusion criteria, Termination criteria, Review and Approval process for Clinical Study, Clinical research protocol		
	(b) Regulatory aspects related to clinical trials		
	• ICH GCP. Quality assurance in Clinical research		

3.	**Biostatistics**	**10**	**33.3%**
	• Definition, application, sample size, importance of sample size, factors influencing sample size, dropouts, statistical tests of significance, type of significance tests, parametric tests (students-t test, ANOVA, Correlation coefficient, regression), non-parametric tests (wilcoxan rank tests, analysis of variance, correlation, chi square test), null hypothesis, P values, degree of freedom, interpretation of P values		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	narrate the drug discovery and development process.
CO2	describe the stages of clinical studies along with relevant guidelines, designing of clinical trial documents including the ethics involved in clinical trials
CO3	enumerate sampling techniques and suggest the sampling technique with justification
CO4	perform statistical analysis of the data generated during pharmaceutical research experiments and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO11
CO1	3	3	3	3	-	-	-	-	-	-	3
CO2	3	3	3	3	-	3	3	3	-	-	3
CO3	3	3	3	3	-	3	3	3	-	-	3
CO4	3	3	3	3	-	-	-	-	-	-	3

Recommended Study Material:

- **References:**

1. New Drug Approval process, 4th Edition, R.A.Guarino, Marcel Dekker, New York.
2. ICH GCP guidelines
3. Schedule Y of Drugs and Cosmetics act
4. Stanford Bolton, Charles Bon (2004), Pharmaceutical Statistics, Practical and Clinical Applications (Fourth rev. ed) Marcel Dekker, Inc.

FACULTY OF PHARMACY
Master of Pharmacy Programme
PHPQA003: Pharmaceutical Quality Assurance Practical-I (Practical)

Total hours: 180 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Analysis of Pharmacopoeial compounds in bulk and in their formulations (tablet/ capsules/ semisolids) by UV Vis spectrophotometer
2	Simultaneous estimation of multi-drug component containing formulations by UV spectrophotometry
3	Experiments based on HPLC
4	Experiments based on Gas Chromatography
5	Estimation of riboflavin/quinine sulphate by fluorimetry
6	Estimation of sodium/potassium by flame photometry or AAS Corrective
7	Case Studies on: • Total Quality Management • Six Sigma • Change Management/ Change control. Deviations, & Deviations • Out of Specifications (OOS) • Out of Trend (OOT) • Corrective & Preventive Actions (CAPA) • Deviations
8	Development of Stability study protocol
9	Estimation of process capability
10	In process and finished product quality control tests for tablets, capsules, parenteral and semisolid dosage forms
11	Assay of raw materials as per official monographs
12	Testing of related and foreign substances in drugs and raw materials
13	To carry out pre formulation study for tablets, parenteral (2 experiment)
14	To study the effect of pH on the solubility of drugs, (1 experiment)

15	Quality control tests for Primary and secondary packaging materials
16	Accelerated stability studies (1 experiment)
17	Improved solubility of drugs using surfactant systems (1 experiment)
18	Improved solubility of drugs using co-solvency method (1 experiment)
19	Determination of pKa and Log P of drugs

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	estimate the pharmaceuticals compounds form bulk and its formulation by UV-Vis spectrophotometry and fluorimetry
CO2	estimate multi-drug component containing formulations by UV spectrophotometry simultaneously.
CO3	separate the components of pharmaceutical mixture by various chromatographic techniques
CO4	perform in process and finished product quality control tests for tablets, capsules, parenterals and semisolid dosage forms
CO5	analyze the problem, communicate suggested solution and interpret the results.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	3	-	-	3	3	3	3	3	3	-
CO2	-	3	-	-	3	3	3	3	3	3	-
CO3	-	3	-	-	3	3	3	3	3	3	-
CO4	-	3	-	-	3	3	3	3	3	3	-
CO5	-	3	-	-	3	3	3	3	3	3	-

Recommended Study Material:

❖ **Textbook:**

1. Principles of Instrumental Analysis, Skoog, Hollar and Nieman, Saunders College Publishers, Philadelphia.
2. Instrumental Methods of Analysis, Willard, Merritt, Dean and Settle, CBS publishers and Distributers, Delhi.
3. Instrumental Methods of Chemical Analysis, G. W. Ewing, McGraw Hill Book Co, NY.

4. Instrumental Methods of Chemical Analysis, B.K. Sharma, Goel Publication House, Meerut, India.
5. John H. Kennedy, Principles of Analytical Chemistry, 2nd Edition, Saunders College Publishing, New York.
6. Higuchi, Bechmman and Hassan: Pharmaceutical Analysis, John Wiley and Sons, New York.
7. D. C. Garratt, The Quantitative Analysis of Drugs, CBS Publishers, New Delhi.
8. P. D. Sethi, Quantitative Analysis of Drugs in Pharmaceutical Formulation, CRC press.
9. J. W. Munson, Pharmaceutical Analysis – Modern Methods, Part – A & B, 2001.
10. United States Pharmacopoeia, United State of Pharmacopeal convention.
11. British Pharmacopoeia, 2010, The British Pharmacopoeia commission office.
12. Indian Pharmacopoeia-2007, Indian pharmacopoeia commission.
13. ICH Q2 R1: Validation of analytical procedures: text and methodology, EMEA, June 25.
14. FDA guidelines For Validation of analytical Method.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS704 B: ACADEMIC SPEAKING (Sem-I)

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/ Week	Theory		Practical		Total
					Internal	External	Internal	External	
1	HS704 A/B/C/D/E/F/G/H	Academic Speaking	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	**Foundations of Advance Communication** • *Meaning and Definition of Advance Communication* • *Advance Communication in Digital, Social, Mobile World* • *Strategies for Advance Communication* • Meaning and Concept of Academic Language • High Frequency Academic Vocabulary	**04**
2	**Art of Conversation** • *Describing people, places and things* • *Expressing opinions* • *Making suggesting* • *Persuading someone* • *Interpreting and Summarizing*	**06**
3	***Science of Power Speaking*** • *Phonemes* • *Word Stress*	**06**

	• *Pronunciation* • *Intonation* • *Pause* • *Register* • *Fluency* • *Prosody* • *Lexical Range*	
4	***Academic Speaking Application – Part I*** • *Art of Oratory* • *Formal Presentation* • *Speech Analysis – Decoding Best Speeches*	**08**
5	**Academic Speaking Application – Part II** • *Job Interview* • *Group Discussion* • *Meeting*	**06**
	Total	**30**

III. **Instruction Methods and Pedagogy**

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, group work, independent and collaborative research, presentations etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	I-Talk	1	10	25
2	Situational Speaking	1	05	
3	Case Study - Speech Analysis	2	10	
4	Attendance and Class Participation	-		05
			Total	**30**

External Evaluation

The University Practical Examination will be for 70 marks and will test the advance communication skills and academic speaking.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course the student would:

CO1	understand and demonstrate advance communication skills and academic speaking.
CO2	demonstrate linguistic competence
CO3	demonstrate performing ability at group discussion and personal interview.
CO4	demonstrate the formal presentation skills.
CO5	demonstrate ability to communicate in diverse situations

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	3	-	-	-
CO2	-	-	-	-	-	-	-	3	-	-	-
CO3	-	-	-	-	3	-	-	3	-	-	-
CO4	-	-	-	-	-	-	-	3	-	-	-
CO5	-	2	-	-	3	-	-	3	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books

- Headway Academic Skills - Level 1: Listening, Speaking and Study Skills Student's Book Paperback

VI. Reading

- **Unit 1:** Business communication Today (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina
- **Unit 2:** Effective Speaking Skills by Terry O' Brien
- **Unit 2:** Speak Better Write Better by Norman Lewis
- **Unit 2:** Well Spoken: Teaching Speaking to All Students by Erik Palmer

- **Unit 3:** Let Us Hear Them Speak : Developing Speaking – Listening Skills in English by Jayshree Mohanraj (Publisher – Sage Publication)
- **Unit 4:** The craft of scientific presentations: Critical steps to succeed and critical errors to avoid. New York: Springer by Michael Alley
- **Unit 4:** Presentation Skills in English by Bob Dignen (Publisher: Orient Black Swan)

FACULTY OF PHARMACY
Master of Pharmacy Programme

Syllabi
Semester II

Charotar University of Science and Technology

FACULTY OF PHARMACY
Master of Pharmacy Programme
PHPQA005: Product Development & Technology Transfer (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Principal of drug discovery and development	12
2	Pre formulation study	12
3	Pilot plant scale up	12
4	Pharmaceutical packaging	12
5	Technology transfer	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Principal of drug discovery and development**	**12**	**20%**
	Introduction, Clinical research process. Development and informational content for Investigational New Drugs Application (IND), New Drug Application (NDA), Abbreviated New Drug Application (ANDA), Supplemental New Drug Application (SNDA), Scale Up Post Approval Changes (SUPAC) and Bulk active chemical Post approval changes (BACPAC), Post marketing surveillance, Product registration guidelines – CDSCO, USFDA. Submission documents for regulators DMFs, as Common Technical Document and Electronic Common Technical Documentation (CTD, eCTD). Concept of regulated and non-regulated markets.		
2.	**Pre formulation study**	**12**	**20%**
	Introduction/concept, organoleptic properties, purity, impurity profiles, particle size, shape and surface area. Solubility, Methods to improve solubility of Drugs: Surfactants & its importance, co-solvency. Techniques for the study of Crystal properties and polymorphism. Pre-formulation protocol, Stability testing during product development.		
3.	**Pilot plant scale up**	**12**	**20%**
	Concept, Significance, design, layout of pilot plant scale up study, operations, large scale manufacturing techniques (formula, equipment, process, stability and quality control) of solids, liquids,		

	semisolid and parenteral dosage forms. New era of drug products: opportunities and challenges.		
4.	**Pharmaceutical packaging**	**12**	**20%**
	Pharmaceutical dosage form and their packaging requirements, Pharmaceutical packaging materials, Medical device packaging, Enteral Packaging, Aseptic packaging systems, Container closure systems, Issues facing modern drug packaging, Selection and evaluation of Pharmaceutical packaging materials. Quality control test: Containers, closures and secondary packing materials.		
5.	**Technology transfer**	**12**	**20%**
	Development of technology by R & D, Technology transfer from R & D to production, Optimization and Production, Qualitative and quantitative technology models. Documentation in technology transfer: Development report, technology transfer plan and Exhibit.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	narrate the new product development process with significance.
CO2	describe scale up activities and justify the operations.
CO3	summarize the packaging requirements for different pharmaceutical dosage forms.
CO4	justify pre-formulation studies in pharmaceutical product development.
CO5	apply concept of technology transfer in pharmaceutical development.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3
CO5	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ **Text book:**

1. The process of new drug discovery and development. I and II Edition (2006) by Charles G. Smith, James T and O. Donnell. CRC Press, Group of Taylor and Francis.

2. Leon Lac Lachman, Herbert A. Liberman, Theory and Practice of Industrial Pharmacy. Marcel Dekker Inc. New York.
3. Sidney H Willing, Murray M, Tuckerman. Williams Hitchings IV, Good manufacturing of pharmaceuticals (A Plan for total quality control) 3rd Edition. Bhalani publishing house Mumbai.
4. Tablets Vol. I, II, III by Leon Lachman, Herbert A. Liberman, Joseph B. Schwartz, 2nd Edn. (1989) Marcel Dekker Inc. New York Anatomy and Physiology in Health and Illness by Kathleen J.W. Wilson, Churchill, Livingstone, New York.
5. Text book of Bio- Pharmaceutics and clinical Pharmacokinetics by Milo Gibaldi, 3rd Edn, Lea & Febriger, Philadelphia.
6. Pharmaceutical product development. Vandana V. Patrevale. John I. Disouza. Maharukh T.Rustomji. CRC Press, Group of Taylor and Francis.
7. Dissolution, Bioavailability and Bio-Equivalence by Abdou H.M, Mack Publishing company, Eastern Pennsylvania.
8. Remingtons Pharmaceutical Sciences, by Alfonso & Gennaro, 19th Edn.(1995)OO2C Lippincott; Williams and Wilkins A Wolters Kluwer Company, Philadelphia.
9. The Pharmaceutical Sciences; the Pharma Path way ‘Pure and applied Pharmacy’ by D. A Sawant, Pragathi Books Pvt. Ltd.
10. Pharmaceutical Packaging technology by D.A. Dean. E.R. Evans, I.H. Hall. 1st Edition (Reprint 2006). Taylor and Francis. London and New York.

FACULTY OF PHARMACY
Master of Pharmacy Programme
PHCCC008: Pharmaceutical Validation and Documentation (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	(a) Introduction of validation (b) Qualification	12
2	(a) Qualification of manufacturing equipment (b) Qualification of laboratory equipment (c) Validation of Utility systems	12
3	Process validation and Analytical method validation	12
4	(a) Cleaning validation (b) Computerized system validation	12
5	Pharmaceutical Documentation	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**(a) Introduction of validation**	**12**	**20%**
	• Definition of Calibration, Qualification and Validation, Scope, frequency and importance. Difference between calibration and validation. Calibration of weights and measures. Advantages of Validation, scope of Validation, Organization for Validation, Validation Master plan, Types of Validation, Streamlining of qualification & Validation process and Validation Master Plan.		
	(b) Qualification		
	• User requirement specification, Design qualification, Factory Acceptance Test (FAT)/Site Acceptance Test (SAT),		

	Installation qualification, Operational qualification, Performance qualification, Re-Qualification (Maintaining status-Calibration Preventive Maintenance, Change management).		
2.	**(a) Qualification of manufacturing equipment**	**12**	**20%**
	• Dry Powder Mixers, Fluid Bed and Tray dryers, Tablet Compression (Machine), Dry heat sterilization/Tunnels, Autoclaves, Membrane filtration, Capsule filling machine.		
	(b) Qualification of laboratory equipment		
	• Hardness tester, Friability test apparatus, tap density tester, Disintegration tester, Dissolution test apparatus		
	(c) Validation of Utility systems		
	• Pharmaceutical water system & pure steam, HVAC system, Compressed air and nitrogen		
3.	**Process Validation**	**12**	**20%**
	• Concept, Process and documentation of Process Validation. Prospective, Concurrent & Retrospective Validation, re validation criteria, Process Validation of various formulations (Coated tablets, Capsules, Ointment/Creams, Liquid Orals and aerosols.), Aseptic filling: Media fill validation, USFDA guidelines on Process Validation- A life cycle approach.		
4.	**(a) Cleaning Validation**	**12**	**20%**
	• Cleaning Method development, Validation of analytical method used in cleaning, Cleaning of Equipment, Cleaning of Facilities. Cleaning in place (CIP). Validation of facilities in sterile and non-sterile plant.		
	(b) Computerized system validation		
	• Electronic records and digital signature - 21 CFR Part 11 and GAMP		
5.	**Pharmaceutical Documentation**	**12**	**20%**

	• Brief introduction to Good Documentation Practices in Pharmaceutical Industry, Documentation control, Paperless documentation system • i) Manufacturing documents: Batch manufacturing record, Batch formula record, Master Formula Record. ii) Quality Assurance Documents: Validation protocols, Validation master plan, analytical method validation protocols, and Validation reports: IQ, OQ, PQ protocols, SOP, Audit Records, and Security Records. iii) Testing Documents: Certificate of Analysis, IPQC protocols, Test method protocol, finished product and raw material protocol, Stability studies protocol iv) Maintenance and Environmental Control related documents. v) Store Management Records: Store reconciliation records for raw material, finished products and Packaging materials. vi) Consumer Related Documents: Product recall, complaint traceability, printing packing.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	describe the concept and preparation aspect of validation and qualification in pharmaceutical industry.
CO2	differentiate qualification and validation process and narrate the process for validating various pharmaceutical instruments and equipment.
CO3	summarize the process validation for pharmaceutical dosage forms.
CO4	narrate the implementation of digital processes and cleaning process in pharmaceutical industry.
CO5	prepare different types of pharmaceutical documents and suggest the approaches for storage of the documents with reference to easy retrieval.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO11
CO1	3	3	3	3	-	-	-	-	-	-	3
CO2	3	3	3	-	-	-	-	-	-	-	3
CO3	3	3	3	-	-	-	-	-	-	-	3
CO4	3	3	3	3	-	3	-	-	-	-	3
CO5	3	3	3	3	-	3	3	-	-	-	3

Recommended Study Material:

❖ **References:**

1. T. Loftus & R. A. Nash, "Pharmaceutical Process Validation", Drugs and Pharm Sci. Series, Vol. 129, 3rd Ed., Marcel Dekker Inc., N.Y.
2. The Theory & Practice of Industrial Pharmacy, 3rd edition, Leon Lachman, Herbert A. Lieberman, Joseph. L. Karig, Varghese Publishing House, Bombay.
3. Validation Master Plan by Terveeks or Deeks, Davis Harwood International publishing.
4. Validation of Aseptic Pharmaceutical Processes, 2nd Edition, by Carleton & Agalloco, (Marcel Dekker).
5. Michael Levin, Pharmaceutical Process Scale-Up‖, Drugs and Pharm. Sci. Series, Vol. 157,2nd Ed., Marcel Dekker Inc., N.Y.
6. Validation Standard Operating Procedures: A Step by Step Guide for Achieving Compliance in the Pharmaceutical, Medical Device, and Biotech Industries, Syed Imtiaz Haider.
7. Pharmaceutical Equipment Validation: The Ultimate Qualification Handbook, Phillip A. Cloud, Interpharm Press.
8. Validation of Pharmaceutical Processes: Sterile Products, Frederick J. Carlton (Ed.) and James Agalloco (Ed.), Marcel Dekker.
9. Analytical Method validation and Instrument Performance Verification by Churg Chan, Heiman Lam, Y.C. Lee, Yue. Zhang, Wiley Interscience.
10. Huber L. Validation and Qualification in Analytical Laboratories. Informa Healthcare.
11. Wingate G. Validating Corporate Computer Systems: Good IT Practice for Pharmaceutical Manufacturers. Interpharm Press.

12. LeBlanc DA. Validated Cleaning Technologies for Pharmaceutical Manufacturing. Interpharm Press.

FACULTY OF PHARMACY
Master of Pharmacy Programme
PHPQA006: Audit & Regulatory Compliance (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction	12
	Role of quality systems and audits in pharmaceutical manufacturing environment	
2	Auditing of vendors and production department	12
3	Auditing of microbiology laboratory	12
	Auditing of Quality Assurance and engineering	
4	Analytical method development and validation	12
5	Bioanalytical method development and validation	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Introduction, Role of quality systems and audits in pharmaceutical manufacturing environment**	**12**	**20%**
	Introduction: Objectives, Management of audit, Responsibilities, Planning process, information gathering, administration, Classifications of deficiencies Role of quality systems and audits in pharmaceutical manufacturing environment: cGMP Regulations, Quality assurance functions, Quality systems approach, Management responsibilities, Resource, Manufacturing operations, Evaluation activities, Transitioning to quality system approach, Audit checklist for drug industries.		
2.	**Auditing of vendors and production department**	**12**	**20%**
	Bulk Pharmaceutical Chemicals and packaging material Vendor audit, Warehouse and weighing, Dry Production: Granulation, tableting, coating, capsules, sterile production and packaging.		
3.	**Auditing of microbiology laboratory, Auditing of Quality Assurance and engineering**	**12**	**20%**
	Auditing of Microbiological laboratory: Auditing the manufacturing process, Product and process information, General areas of interest in the building raw materials, Water, Packaging		

	materials. Auditing of Quality Assurance and engineering department: Quality Assurance Maintenance, Critical systems: HVAC, Water, Water for Injection systems, ETP.		
4.	**Analytical method development and validation**	**12**	**20%**
	Case studies for analytical method development (Spectroscopic and chromatographic methods) Need of analytical method validation, Validation parameters: definition, methodology and acceptance criteria, Statistical considerations in Analytical method validation and application of experimental designs, Comparative study of guidelines as per ICH.		
5.	**Bioanalytical method development and validation**	**12**	**20%**
	Importance of Bioanalytical method development, Critical optimization steps for method development. Strategies, planning and workflow for bioanalytical method validation, Regulatory guidelines for BMV. Differentiating steps from Analytical Method Validation.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	narrate the roles of quality system and audits in pharmaceutical environment and propose the steps of audits with significance.
CO2	describe the audit process for vendors and production department
CO3	prepare the auditing report and the check list for auditing
CO4	describe analytical method development and validation process on line with various guidelines proposed by competent authorties
CO5	describe bioanalytical method development and validation process with significance of various processes.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3
CO5	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

- ❖ **Text book:**
  1. Compliance auditing for Pharmaceutical Manufacturers. Karen Ginsbury and Gil Bismuth, Interpharm/CRC, Boca Raton, London New York, Washington D.C.
  2. Pharmaceutical Manufacturing Handbook, Regulations and Quality by Shayne Cox Gad. WileyInterscience, A John Wiley and sons, Inc., Publications.
  3. Handbook of microbiological Quality control. Rosamund M. Baird, Norman A. Hodges, Stephen P. Denyar. CRC Press. 2000.
  4. Laboratory auditing for quality and regulatory compliance. Donald C. Singer, Raluca-loana Stefan, Jacobus F. Van Staden. Taylor and Francis (2005).
  5. ICH guideline for method validation.
  6. EMEA for Bioanalytical method validation.
  7. Industrial guidelines for Bioanalytical method validation.

FACULTY OF PHARMACY

Master of Pharmacy Programme

PHPQA007: Pharmaceutical Manufacturing Technology (Theory)

Total hours: 60 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Pharmaceutical industry developments	12
	Plant layout	
2	Aseptic process technology	12
3	Non sterile manufacturing process technology	12
4	Containers and closures for pharmaceuticals	12
5	Quality by design (QbD) and process analytical technology (PAT)	12

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Pharmaceutical industry developments, Plant layout**	**12**	**20%**
	Pharmaceutical industry developments: Legal requirements and Licenses for API and formulation industry, Plant location- Factors influencing. Plant layout: Factors influencing, Special provisions, Storage space requirements, sterile and aseptic area layout. Production planning: General principles, production systems, calculation of standard cost, process planning, routing, loading, scheduling, dispatching of records, production control.		
2.	**Aseptic process technology**	**12**	**20%**
	Manufacturing, manufacturing flowcharts, in process quality control tests for following sterile dosage forms: Ointment, Suspension and Emulsion, Dry powder, Solution (Small Volume & large Volume). Advanced sterile product manufacturing technology: Area planning & environmental control, wall and floor treatment, fixtures and machineries, change rooms, personnel flow, utilities & utilities equipment location, engineering and maintenance. Process Automation in Pharmaceutical Industry: With specific reference to manufacturing of sterile semisolids, Small Volume Parenterals & Large Volume Parenterals (SVP & LVP), Monitoring of Parenteral manufacturing facility, Cleaning in Place (CIP), Sterilization in Place (SIP), Prefilled Syringe,		

	Powdered Jet, Needle Free Injections, and Form Fill Seal Technology (FFS). Lyophilization technology: Principles, process, equipment.		
3.	**Non sterile manufacturing process technology**	**12**	**20%**
	Manufacturing, manufacturing flowcharts, in process-quality control tests for following non-Sterile solid dosage forms: Tablets (compressed & coated), Capsules (Hard & Soft). Advance non-sterile solid product manufacturing technology: Process Automation in Pharmaceutical Industry with specific reference to manufacturing of tablets and coated products, Improved Tablet Production: Tablet production process, granulation and pelletization equipments, continuous and batch mixing, rapid mixing granulators, rota granulators, spheronizers and marumerisers, and other specialized granulation and drying equipments. Problems encountered. Coating technology: Process, equipments, particle coating, fluidized bed coating, application techniques. Problems encountered.		
4.	**Containers and closures for pharmaceuticals**	**12**	**20%**
	Types, performance, assuring quality of glass; types of plastics used, Drug plastic interactions, biological tests, modification of plastics by drugs; different types of closures and closure liners; film wrapper; blister packs; bubble packs; shrink packaging; foil / plastic pouches, bottle seals, tape seals, breakable seals and sealed tubes; quality control of packaging material and filling equipment, flexible packaging, product package compatibility, transit worthiness of package, Stability aspects of packaging. Evaluation of stability of packaging material.		
5.	**Quality by design (QbD) and process analytical technology (PAT)**	**12**	**20%**
	Current approach and its limitations. Why QbD is required, Advantages, Elements of QbD, Terminology: QTPP. CMA, CQA, CPP, RLD, Design space, Design of Experiments, Risk Assessment and mitigation/minimization. Quality by Design, Formulations by Design, QbD for drug products, QbD for Drug Substances, QbD for Excipients, Analytical QbD. FDA initiative on process analytical technology. PAT as a driver for improving quality and reducing costs: quality by design (QbD), QA, QC and GAMP. PAT guidance, standards and regulatory requirements.		

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	narrate pharmaceutical industry developments, plant layout and production planning
CO2	describe the principles and practices of aseptic process technology, non-sterile manufacturing technology and packaging technology with justification for adaptation.
CO3	summarize the principles of packaging and suggest various packages for pharmaceuticals with justifications.
CO4	describe principles of Quality by design (QbD) and process analytical technology (PAT) in pharmaceutical manufacturing and apply those principles in prototype conditions.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	3	-	3
CO3	3	-	-	-	-	-	-	-	3	-	3
CO4	3	-	-	3	-	-	-	-	-	-	3

Recommended Study Material:

❖ **Textbook:**

1. Lachman L, Lieberman HA, Kanig JL. The theory and practice of industrial pharmacy, 3rd ed., Varghese Publishers, Mumbai 1991.
2. Sinko PJ. Martin's physical pharmacy and pharmaceutical sciences, 5th ed., B.I. Publications Pvt. Ltd, Noida, 2006.
3. Lieberman HA, Lachman L, Schwartz JB. Pharmaceutical dosage forms: tablets Vol. I-III, 2nd ed., CBS Publishers & distributors, New Delhi, 2005.
4. Banker GS, Rhodes CT. Modern Pharmaceutics, 4th Ed. Inc, New York, 2005. Marcel Dekker
5. Sidney H Willing, Murray M, Tuckerman. Williams Hitchings IV, Good manufacturing of pharmaceuticals (A Plan for total quality control) 3rd Edition. Bhalani publishing house Mumbai.
6. Indian Pharmacopoeia. Controller of Publication. Delhi, 1996.
7. British Pharmacopoeia. British Pharmacopoeia Commission Office, London, 2008.
8. United States Pharmacopoeia. United States Pharmacopeial Convention, Inc, USA, 2003.

9. Dean D A, Evans E R and Hall I H. Pharmaceutical Packaging Technology. London, Taylor & Francis, 1st Edition. UK.
10. Edward J Bauer. Pharmaceutical Packaging Handbook. 2009. Informa Health care USA Inc. New york.
11. Shaybe Cox Gad. Pharmaceutical Manufacturing Handbook. John Willey and Sons, New Jersey, 2008.

FACULTY OF PHARMACY
Master of Pharmacy Programme
PHCCC004: Research Methodology (Theory)

Total hours: 30 Hr

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	(c) General Research Methodology	10
	(d) Declaration of Helsinki	
2	(a) Medical Research	10
	(b) CPCSEA guidelines for laboratory animal facility	
3	Design of Experiments	10

Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**(c) General Research Methodology**	**10**	**33.33%**
	• Research, objective, requirements, practical difficulties, review of literature, study design, types of studies, strategies to eliminate errors/bias, controls, randomization, crossover design, placebo, blinding techniques.		
	(d) Declaration of Helsinki		
	• History, introduction, basic principles for all medical research.		
2.	**(a) Medical Research**	**10**	**33.33%**
	• History, values in medical ethics, guidelines, ethics committees, conflicts of interests, referral, treatment of family members, fatality.		
	(b) CPCSEA guidelines for laboratory animal facility		
	• General introduction, guidelines applicable in India, constitution and function of the committee, record keeping,		

	SOPs, personnel and training, transport of lab animals		
3.	**Design of Experiments**	**10**	**33.33%**
	• Variables - Definition of Variable, Types of variables (Dependent and Independent variables, confounded variables), Measurement of variables, Types of measurement scales and their comparison. Reliability and Validity of Measurements. • Definition, Need, Advantages of DOE in Pharmacoinformatics Experimental designing, planning of an experiment, description of terms such as replication and randomization. • Placket-Burman design, Factorial-Fractional designs, Simplex design, mixture designs, central composite designs and Box Benken method in optimization, Application of optimization techniques in QbD in product development Evolution of full and reduced mathematical models in experimental designs, polynomial equations, Response surface methodology, contour plots, and their interpretation. • Application of DOE in pharmaceutical research: Discipline specific applications.		

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	narrate hierarchy of continue research by proper fundamental methodology.
CO2	summarize the guidelines and ethical values in medical research.
CO3	prepare protocol for Animal study.
CO4	apply the concept of design of experiments in pharmaceutical research.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	3	-	-	3

Recommended Study Material:

❖ **References:**

1. Research In Education- John V. Best, John V. Kahn 7th edition, Published by: Phi Learning Pvt. Ltd
2. Research Methodology: Methods and Techniques. C.R. Kothari and Gaurav Garg, New Age International Publications.
3. Essentials of Research Methodology and Dissertation Writing. Kanan Yelikar, Jaypee Publishers
4. ICMR Ethical Guidelines for Biomedical Research (http://icmr.nic.in/ethical_guidelines.pdf)
5. A review of ―Scientist in legal Systems‖, Journal of Forensic Sciences (JOFS),21(2),1976.
6. Donald Menzel, Jones, Howard Mumford; Boyd, Lyle G., Writing a technical paper, J. Chem. Edu., 1962, 39 (6), p A500.
7. Mitchell, K. and Glover, T. 2001. Introduction to Biostatistics. McGraw Hill, Publishing Co.

FACULTY OF PHARMACY
Master of Pharmacy Programme
PHPQA008: Pharmaceutical Quality Assurance Practical-II (Practical)

Total hours: 180 Hr

Outline of the course:

Sr. No.	Title of the unit
1	Organic contaminants residue analysis by HPLC
2	Estimation of Metallic contaminants by Flame photometer
3	Identification of antibiotic residue by TLC
4	Estimation of Hydrogen Sulphide in Air.
5	Estimation of Chlorine in Work Environment.
6	Sampling and analysis of SO2 using Colorimetric method
7	Qualification of following Pharma • Autoclave • Hot air Oven • Powder Mixer (Dry) • Tablet Compression Machine
8	Validation of an analytical method for a drug
9	Validation of a processing area
10	Qualification of at least two analytical instruments
11	Cleaning validation of one equipment
12	Qualification of Pharmaceutical Testing Equipment (Dissolution testing apparatus, Friability Apparatus, Disintegration Tester)
13	Check list for Bulk Pharmaceutical Chemicals vendors
14	Check list for tableting production
15	Check list for sterile production area
16	Check list for water for injection
17	Design of plant layout: Sterile and non-sterile
18	Case study on application of QbD

19	Case study on application of PAT

Course Outcome (COs):

At the end of the course, the students would be able to;

CO1	summarize the concept of pharmaceutical validation and carry out validation of selected equipment/instruments.
CO2	apply basic instrumental techniques in pharmaceutical analysis for assessing the quality of formulations.
CO3	apply routine statistical tools in field of pharmaceutical analysis.
CO4	qualify pharmaceutical testing instruments and manufacturing equipment.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	3	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3

Recommended Study Material:

❖ **Textbook:**

1. Principles of Instrumental Analysis, Skoog, Hollar and Nieman, Saunders College Publishers, Philadelphia.
2. Instrumental Methods of Analysis, Willard, Merritt, Dean and Settle, CBS publishers and Distributers, Delhi.
3. Instrumental Methods of Chemical Analysis, G. W. Ewing, McGraw Hill Book Co, NY.
4. Instrumental Methods of Chemical Analysis, B.K. Sharma, Goel Publication House, Meerut, India.
5. John H. Kennedy, Principles of Analytical Chemistry, 2nd Edition, Saunders College Publishing, New York.
6. Higuchi, Bechmman and Hassan : Pharmaceutical Analysis, John Wiley and Sons, New York.
7. D. C. Garratt, The Quantitative Analysis of Drugs, CBS Publishers, New Delhi.
8. P. D. Sethi, Quantitative Analysis of Drugs in Pharmaceutical Formulation, CRC press.
9. J. W. Munson, Pharmaceutical Analysis – Modern Methods, Part – A & B, 2001.
10. United States Pharmacopoeia, United State of Pharmacopeal convention.
11. British Pharmacopoeia, 2010, The British Pharmacopoeia commission office.

12. Indian Pharmacopoeia-2007, Indian pharmacopoeia commission.
13. ICH Q2 R1: Validation of analytical procedures: text and methodology, EMEA, June 25.
14. FDA guidelines For Validation of analytical Method.
15. WHO guidelines For Validation of analytical Method.
16. How to perform GMP, P P Sharma, Vandana Publications, New Delhi.
17. Handbook of cosmetics: Formulation, Manufacturing and quality control, by PP sharma, Vandana Publications, New Delhi.
18. Pearson’s Composition and analysis offoods, R.S.Kirk, R.sawyer, Addson Weskey, England.
19. Handbook of Forensic analysis, Fedrick P, Academic Press.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS705 (B): ACADEMIC WRITING

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
II	HS705 (B)	Academic Writing	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title / Topic	Classroom Contact Hours
1	**Academic Writing and Research Process** • *Introduction to Academic Writing* • *Academic Writing as a Part of Research* • *Types of Academic Writing* • *Features of Academic Writing* • *Importance of Good Academic Writing in various Academic Works*	05
2	**Anatomy of Academic Writing** • *Academic Vocabulary* • *Simple and Complex Sentences* • *Organizing Paragraphs* • *The Writing Process* • *Adopting Academic Writing Style*	05
3	**Key Academic Skills** • Note – taking • Note – making	05

	• Paraphrasing • Summarizing	
4	**Accuracy in Academic Writing** • *Lexical Range* • *Academic Language and Structures* • *Elements of Writing* • *Proof Reading, Editing, and Rewriting*	**05**
5	**Using and Citing Sources of Ideas** • *Academic Texts and their Types* • *Intellectual Honesty in Academic Writing* • *Avoiding Plagiarism – Idea Theft* • *Degrees of Plagiarism* • *Types of Borrowing* • *Anatomy of Citations* • *Common Citation Styles*	**05**
6	**Contemporary Practices in Academic Writing** • Analytical Essays • Graph / Table / Process Interpretation and Description • Writing Reports • Writing Research / Concept Papers	**05**
Total		**30**

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like writing, group work, independent and collaborative research, etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Paragraph Writing	1	3	03
2	Note-taking / Note-making	1	3	03
3	Paraphrasing / Summarizing	1	4	04
4	Essay Writing	1	5	05
5	Concept Paper Writing	1	10	10
5	Attendance and Class Participation			05
			Total	**30**

External Evaluation

The University Practical Examination will be for 70 marks and will test the professional communication skills and academic writing skills of the students.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical /Quiz/ Project / Academic Writing	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course, the student would:

CO1	have sound understanding of the concept and applications of academic writing
CO2	have acquired enough knowledge of academic writing style, strategy and approach
CO3	be able to demonstrate error free and effective academic writing
CO4	be able to demonstrate ability to work on project/report/paper writing
CO5	understand the concept of plagiarism and learn to use different citation styles as a part of referencing

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	3	-	-	-
CO2	-	-	-	-	-	-	-	3	-	-	-
CO3	-	-	-	-	-	-	-	3	-	-	-
CO4	-	-	-	-	-	-	-	3	-	-	-
CO5	-	-	-	-	-	-	2	-	-	-	-

Course Articulation Matrix:

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books / Reading

Essential Reading for Concepts

- Academic Writing for International Students, Routledge
- Academic Writing: A Guide for Management Students and Researchers. Monipally, M. M. & Pawar, B. S. Sage. 2010. New Delhi

Essential Reading for Activity and Teacher Resource

- ***Effective Academic Writing Level - 1,2,3,4 (Second Edition) By:*** Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; *Publisher: Oxford*

Additional Reading

- Writing Your Thesis (2nd Edition) by Paul Oliver, Sage
- Development Communication In Practice by Vilanilam V J, Sage
- Intercultural Communication by Mingsheng Li, Patel Fay, Sage
- www.owl.perdue.edu.

FACULTY OF PHARMACY
Master of Pharmacy Programme

Syllabi
Semester III

Charotar University of Science and Technology

FACULTY OF PHARMACY
Master of Pharmacy Programme
PHPQA010: Dissertation Part-I

Total hours: 480

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	define and describe research problem.
CO2	illustrate project management skills such as project design, scientific information and literature access, project implementation, data analysis, and interpretation.
CO3	present a dissertation report integrating appropriate written and verbal communicative skills.
CO4	efficiently use communication and information technology tools.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	3	3	3	-	-	-	-	-	-	3
CO3	3	-	3	3	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	3	3	-	-	3

FACULTY OF PHARMACY
Master of Pharmacy Programme
PHCCC005: Journal Club -I

Total hours: 30

Course Outcome (COs):

At the end of the course, the students would be able to

	CO
CO1	present scientific literature and interpret the finding

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	3	-	-	-	3	-	-	3

Master of Pharmacy Programme

Syllabi
Semester IV

Charotar University of Science and Technology

FACULTY OF PHARMACY
Master of Pharmacy Programme
PHPQA011: Dissertation Part-II

Total hours: 480

Course Outcome (COs):

At the end of the course, the students would be able to

CO1	define and describe research problem.
CO2	illustrate project management skills such as project design, scientific information and literature access, project implementation, data analysis, and interpretation.
CO3	present a dissertation report integrating appropriate written and verbal communicative skills.
CO4	efficiently use communication and information technology tools.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	3	3	3	-	-	-	-	-	-	3
CO3	3	-	3	3	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	3	3	-	-	3

FACULTY OF PHARMACY
Master of Pharmacy Programme
PHCCC006: Journal Club -II

Total hours: 30

Course Outcome (COs):

At the end of the course, the students would be able to

	CO
CO1	present scientific literature and interpret the finding

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	3	-	-	-	3	-	-	3

UNIVERSITY ELECTIVE LEVEL-POSTGRADUATE

CHAROTAR UNIVERSITY OF SCIENCE&TECHNOLOGY

List of University Elective Courses

First Semester Postgraduate Programme

(Academic year 2019-20)

Sr. No.	Course Code & Course Name	Department / Faculty offering the course
1	MA771.01: Reliability and Risk Analysis-I	PDPIAS / FAS
2	EE781.01:Optimization Techniques	EE/FTE
3	CE772.01:Research Methodology	CE / FTE
4	CA730: Internet and Web Designing	CMPICA / FCA
5	PT795.01 Heath and Physical activity	ARIP / FMD
6	NR755: First aid and Life support	NURSING / FMD
7	RD701.01: Introduction to analytical Techniques	PDPIAS/FAS
8	RD702.01:Introduction to Nanoscience and Technology	PDPIAS/FAS
9	MB650: Creative Leadership	I2IM / FMS
10	PH891:Community Pharmacy ownership	RPCP/FPH
11	ME781.01: Occupational Health and Safety	ME/FTE
12	PD261: Astrophysics, Space and Cosmos-1(ASC-1)	PDPIAS/FAS

MA771.01: RELIABILITY AND RISK ANALYSIS-I

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	NA	**2**	2	2
Marks	NA	100	100	

A. Outline of the course:

Sr No.	Title of the unit	Minimum number of hours
1.	Introduction to Probability and Statistics	06
2.	Characteristics of Reliability	06
3.	Reliability of Simple Systems	06
4.	Concept of Safety and Risk Analysis	06
5.	Reliability Modeling	06

Total Hours (Theory): 30

B. Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1	**Introduction to Probability and Statistics**	**06 Hours**	**20%**
1.1	Random Event.		
1.2	Basic formula of Probability.		
1.3	Random Variable and Probability Distribution Functions.		
1.4	descriptive statistics		
2	**Characteristics of Reliability**	**06 Hours**	**20%**
2.1	Reliability of a Unit Functioning until First Failure		
2.2	System Reliability		
2.3	Testing for Reliability		
2.4	Exponential Law and Evaluation of parameter		
3.	**Reliability of Simple Systems**	**06 Hours**	**20%**

3.1	Series System	
3.2	Parallel System	
3.3	K out of N systems	
4.	**Concept of Safety and Risk Analysis** **06 Hours**	**20%**
4.1	Qualitative definition of Risk	
4.2	Quantitative definition of Risk	
4.3	Failure Model and Effect Analysis(FMEA)	
4.4	Hazard and operability analysis(HAZOP)	
4.5	Fault Tree Analysis	
5.	**Reliability Modeling** **06 Hours**	**20%**
5.1	Software Reliability Analysis	
5.2	Human Reliability	
5.3	Stress-Strength Analysis	

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures/laboratory which carries a 5% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weighting of 5%.
- Two Quizzes (surprise test) will be conducted which carries 5% component of the overall evaluation.

D. Student Learning Outcomes:

- At the end of the course the students will be able to understand the basic concepts of Reliability and Risk Analysis.

- Student will be able to apply concepts of these course in their study of specialization

E. Recommended Study Material:

❖ Text Books:

1. Mathematical Methods of Reliability Theory. B. V. Gnedenko, Yu. K. Belyayev, and A. D. Solovyev ,Academic Press 1969

2. An Introduction to Basics of Reliability and Risk Analysis. Enric Zio, World Scientific Publishing Co.Pte. Ltd.2007
3. Reliability and Risk Analysis. Terje Aven, Elsevier Publicaion,1992

❖ **Reference Books/Articles:**

1. On The Quantitative Definition of Risk, Stanley Kaplan and B. John Garrick, Risk Analysis, Vol. I, No. I , 1981
2. Probability concepts in engineering planning and design. Volume II – decision, risk and reliability. Ang, A.H.-S. and Tang, W.H John Wiley & Sons, Inc., New York (1984)
3. M. Modarres, Reliability and Risk Analysis, Marcel Dekker (1993).
4. N.J. McCormick, Reliability and Risk Analysis, Academic Press (1981).

EE781.01: OPTIMIZATION TECHNIQUES

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of the Course:

Sr. No.	Title of Unit	Min. no of hours
1	Fundamentals of Optimization	01
2	Linear Programming	05
3	Unconstrained Optimization	04
4	Nonlinear Programming	05
5	Fundamentals of Artificial Intelligence methods	01
6	Particle Swarm Optimization and Cuckoo Search Algorithm	08
7	MATLAB programming and Optimization Toolbox	06

Total Hours (Theory): 30

B. Detailed Syllabus

Sr. No.	UNIT	Hours	Weightage (%)
1	Fundamentals of Optimization	01 Hours	3.33%
	Introduction, Feasibility and optimality, Convexity, constraints, Rates of convergence		
2	Linear programming	05 Hours	16.66%
	Introduction, Formulation of Linear programming problem, Graphical method, Simplex method, Basic solution, Basic feasible solution, Simplex algorithm, Two phase method.		
3	Unconstrained Optimization	04 Hours	13.33%
	Introduction, Optimality conditions, Newton’s method for minimization, Line search methods, Steepest-Descent method, Quasi-Newton method, Modified newton’s method		
4	Nonlinear Programming	05 Hours	16.66%

	Optimality conditions for constrained problems, Kuhn-Tucker conditions, Penalty function method, Barrier method, The Lagrange multipliers and the Lagrangian function, Sensitivity analysis, Computing the Lagrange multipliers, Sequential quadratic programming, Interior point method		
5	Fundamentals of Artificial Intelligence methods	01 Hours	3.33%
	To understand importance of AI methods and their comparison with various classical methods using various criteria.		
6	Particle Swarm Optimization and Cuckoo Search Algorithm	08 Hours	26.66%
	Introduction to PSO, Unconstrained and constrained optimization using PSO, Effects of various coefficients on convergence, Cuckoo Search (CS) Algorithm, Comparison between PSO and CS		
7	MATLAB programming and Optimization Toolbox	06 Hours	20%
	MATLAB programming of various classical and AI methods. Use of MATLAB optimization toolbox to solve various optimization problems.		

C. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a component of the overall evaluation.
- Minimum two internal exams will be conducted and will be considered as a part of overall evaluation.
- Assignments based on course content will be given to the students for each unit/topic and will be evaluated at regular interval and its weightage may be reflected in the overall evaluation.

D. Student Learning Outcomes:

A. The students will be able to get awareness about the optimization problems. They can differentiate the class of classical optimization methods and AI methods

B. The student will learn to handle, solve and analyzing problems using linear programming and other mathematical programming algorithms

C. The students will also be able to learn different techniques to solve Non- Linear Programming Problems. They can also use search techniques methods, which are based on iterative methods, to find optimal solutions of Non-Linear Programming Problems.

D. Ability to develop codes for evolutionary optimization techniques and to solve wide range of optimization problem.

E. The students will be able to solve optimization methods using software tools such as MATLAB and be prepared for developing case studies and simulation examples

E. Recommended Study Material:

Books:

1. "Optimization Methods for Engineers", N.V.S. Raju, PHI, 2014.
2. "Artificial intelligence and intelligent systems", N.P. Padhy, Oxford University Press, 2005
3. "Engineering Optimization: Theory and Practice", S. Rao, 4th Edition, John Wiley & Sons, Inc., 2009

CE772.01: RESEARCH METHODOLOGY

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	**2**	2	2
Marks	NA	100	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	General introduction to Research	02
2	Research problem Formulation	03
3	Research Design	08
4	Research Publication & Presentation	08
5	Research Ethics and Morals	05
6	Quality indices of research publication	04

Total hours: 30

B. Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1	**General introduction to Research**	**02 Hours**	**06%**
	General Introduction: Importance of Research, Role of Research, Aims & Objectives, Research Process, Phases of Research. Introduction to Research Methodology: Meaning of Research, Objectives of Research, Motivations in Research, Types of Research, Research Approaches, Significance of Research, Research Methods v/s Methodology, Research and Scientific Methods, Research Process, Criteria of Good Research.		
2.	**Research problem Formulation:**	**03 Hours**	**09%**

	Review of Research Literature: Defining the Research Problem: What is Research Problem, Selecting the Problem, Necessity of and Techniques in defining the problem. Purpose and use of literature review, locating relevant information, use of library & electronic databases, preparation & presentation of literature review, research article reviews, theoretical models and frame work. Identification of gaps in research, formulation of research problem, definition of research objectives.		
3.	**Research Design:**	**08 Hours**	**27%**
	Research Design: Meaning, Need, Features of Good Design, Concepts, Types. Basic Principles of Experimental Design, Developing a Research Plan. *Qualitative Methods:* Types of hypothesis and characterization. *Quantitative Methods:* Statistical methods for testing and evaluation. *Characterization of experiments:* Accuracy, reliability, reproducibility, sensitivity, Documentation of ongoing research. *Sample Design:* Implication, Steps. Criteria for selecting a sample procedure, Characteristics of Good sampling Procedure, Types of Sample Design, Selecting Random Samples, Complex random sampling Design. *Measurement and Scaling Techniques:* Measurement in Research, Measurement Scales, Sources of Errors in measurement, Tests of Second measurement, Technique of developing Measurement Tools, Meaning of Scaling, Scale Classification Bases, Important Scaling Techniques, Scale Construction Techniques. *Methods of Data Collection:* Collection of Primary Data, Observation Method, Interview method, Collection of Data through questionnaire and Schedules, Other methods. Collection of Secondary Data, Selection of appropriate method for data collection, Case Study Method, Guidelines for developing questionnaire, successful interviewing. Survey v/s Experiment. *Processing and Analysis of Data :* Processing Operations (Meaning, Problems), Data Analysis (Elements), Statistics in Research, Measures of Central Tendency, Dispersion, Asymmetry, Relationship. Regression Analysis, Multiple		

	correlation and Regression, Partial Correlation, Association in case of Attributes. *Sampling Fundamentals:* Definition, Need, Important sampling Distribution, Central limit theorem Sampling Theory, Sandler's A-test, Concept of Standard Error, Estimation, Estimating population mean, proportion. Sample size and its determination, Determination of sample size. *Analysis of Variance and Covariance:* Basic Principles, techniques, applications, Assumptions, limitations. *Analysis of Non-parametric or distribution-free Tests :* Sign Test, Fisher-Irwin Test, McNemer Test, Wilcoxon Matched pair Test (Signed Rank Test). *Sum Tests :* a) Wilcoxon-Mann-Whitney Test b)Kruskal-Wallis Test, One sample Runs Test, Multivariate Analysis Techniques: Characteristics, Application, Classification, Variables, Techniques, Factor Analysis (Methods, Rotation), Path Analysis.		
4.	**Research Publication & Presentation:**	**08 Hours**	**27%**
	Thesis, Research paper, Organization of thesis and reports, formatting issues, citation methods, references, effective oral presentation of research, Documentation of ongoing research.		
5.	**Research Ethics and Morals:**	**05 Hours**	**19%**
	Issues related to plagiarism and ethics. Intellectual Property Rights: Copy rights, Patents, Industrial Designs, Trademarks.		
6.	**Quality indices of research publication:**	**04 Hours**	**12%**
	Impact factor, Immediacy factor.		

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- Tutorials related to course content will be given to students.

- In the lectures discipline and behaviour will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

D. Student Learning Outcomes:

- Research Methodology as a subject should help researchers to prepare the literature in chronological pattern and should logically analyse the concerns.
- This subject should help in framing the research problems to enhance the scale of understanding.
- In this world of global village, research papers are available in abundance; one thesis submitted by a scholar, in no way should be a repetition of a work already done.
- This subject should help researchers to use tools, techniques, concepts and world's best practices to present a unique research.

E. Recommended Study Material:

❖ **Text Books:**

1. Research Methodology, Methods & Techniques, C.R. Kothari, Viswa Prakashan, 2nd Edition, 2009.
2. Research Methods- A Process of Inquiry, Graziano, A.M., Raulin, M.L, Pearson Publications, 7th Edition, 2009.
3. How to Write a Thesis:, Murray, R. Tata McGraw Hill, 2nd Edition, 2010.
4. Writing For Academic Journals, Murray, R., McGraw Hill International, 2009.
5. Writing for Publication, Henson, K.T., Allyn &Bacon, 2005.

❖ **Reference Books:**

1. What is this thing called Science, Chalmers, A.F., Queensland University Press, 1999.
2. Methods &Techniques of Social Research, Bhandarkar & Wilkinson, Himalaya publications, 2009.
3. Doing your Research project, Bell J., Open University Press, Berkshire, 4thEdition, 2005
4. A Handbook of Academic Writing, Murray, R. and Moore, S., Tata McGraw Hill International, 2006.

CA730: INTERNET AND WEB DESIGNING

Credits and Hours:

Teaching Scheme	**Theory**	**Practical**	**Total**	**Credit**
Hours/week	NA	**2**	2	2
Marks	**NA**	100	100	

A. Outline of the Course:

Sr. No.	**Content**
1.	Overview of Internet and WWW, Basic elements of the Internet, Internet services, Internet Browsers and Servers, Hardware and Software requirements to connect to the internet, Internet Service Provider (ISP), Introduction to Internet Protocols
2.	Introduction to Web Page, Web Site, Web Browser, Overview of HTML, Structure of HTML Documents
3.	HTML Basics Tags and HTML elements
4.	List, Marquee & Hyperlink in HTML
5.	Images and Tables in HTML
6.	Forms in HTML
7.	Media Element in HTML5
8.	Introduction to Cascading Style Sheet (CSS), Ways to embed CSS in HTML
9.	CSS selectors & Layout
10.	CSS Properties
11.	Creation of Menu with CSS
12.	Introduction to Web Publishing or Hosting : Domain Name, Web Server, Website Parking, Publishing Website through FTP

Total Hours (Theory): 30

❖ **Text Books:**

1. Harley Hahn: The Internet Complete Reference, 2nd Edition, Tata McGraw-HILL Edition.
2. Matthew MacDonald: HTML5: The Missing Manual, O'Reilly Media, August 2011.
3. Peter Gasston: The Book of CSS3: A Developer's Guide to the Future of Web Design, No Starch Press, April 2011.
4. Richard York: Beginning CSS: Cascading Style sheets for Web Design, Wrox Press (Wiley Publishing), 2005.

❖ **Reference Books**:

1. Ivan Bayross: Web Enabled Commercial Application Development using HTML, JavaScript, DHTML and PHP, 4th revised edition, BPB Publication.

2. Adrian Farrel: The Internet and its Protocol – A comparative approach, Morgan Kaufmann Publishers.
3. David Mc Farland: CSS: The Missing Manual, O'Reilly, 2006.

❖ **Reference Links**:

1. http://www.w3schools.com [lecture notes]
2. http://www.whatwg.org/specs/web-apps/current-work/multipage/#auto-toc-4 [HTML Materials]
3. http://people.cs.pitt.edu/~mehmud/cs134-2084/lectures.html [CSS notes]

PT795.01: HEALTH & PHYSICAL ACTIVITY

Total Hours (Theory): 30

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to health	10
2.	Physical Activity	10
3.	Introduction to Yoga	10

B. Detailed Syllabus:

1	**Introduction to Health**	**10 hours**
1.1	What is health?	
1.2	Healthcare delivery system: Developing Countries, Developed Countries	
1.3	Human anatomy, physiology & physical fitness	
1.4	Basics of Nutrition	
1.5	Life style disorders – obesity & diabetes	
2	**Physical Activity**	**10 hours**
2.1	What is Physical activity, exercise, physical fitness, epidemiology?	
2.2	Measurement of Physical Activity in individuals	
2.3	Physical Activity – Theoretical Perspective: Self-determination, trans theoretical	
2.4	Physical Activity and mental health – Body image, depression, problem with exercise	
2.5	Barriers & Facilitators of Physical Activity	
3	**Introduction to Yoga**	**10 hours**
3.1	What is yoga?	
3.2	Types of yoga	
3.3	Benefits of yoga to various body systems	
3.4	Asanas, Pranayam	
3.5	Yoga therapy for various back pain, asthma, stress, hypertension, diabetes	

C. Instructional Method and Pedagogy:

- Interactive classroom sessions using black-board and audio-visual aids.
- Using the available technology and resources for e-learning.
- Students will be encouraged towards self-learning and under direct interaction with course faculty.
- Students will be enabled for continuous evaluation.
- Case study, didactic mode of group discussions

D. Student Learning Outcomes:

Upon completion of the course, the student should be able to:

- Appraise the importance of exercise in maintenance of health and fitness.
- Objectively define health and physical activity in realistic environment.

E. Recommended Study Material:

❖ **Textbooks:**

1. ACSM"s "Health Related Physical Fitness Assessment Manual Lippincott Williams and Walkins USA, 2005.
2. Nilima Patel (2008) Yoga and Rehabilitation, Jaypee Publication, India

❖ **Reference books:**

1. Biddle, S. J. H., & Mutrie, N. (2008). Psychology of physical activity. London: Routledge
2. B.C. Rai. Health Education and Hygiene Published by Prakashan Kendra
3. Puri. K. Chandra. S.S. (2005). Health and Physical Education. New Delhi: Surjeet Publications

NR755: First Aid and Life Support

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	**2**	2	2
Marks	NA	100	100	

A. Outline of the Course:

Unit No.	Title of Unit	Prescribed Hours
I.	**Introduction and Basics of First Aid:** □ ● Rescuer Duties, Victim and Rescuer Safety ● Looking for Help □ ● After the emergency	2
II.	**Medical emergencies and their first aid:** □ ● Breathing Problems □ ● Choking in an Adult □ ● Allergic Reactions □ ● Heart Attack ● Fainting □ ● Diabetes and Low Blood Sugar □ ● Stroke □ ● Shock Abdominal maneuver, CPR, ventilation etc.	12
III.	**Injuries emergencies and their first aid:** ● Bleeding: You Can See/ You Can't See ● Wounds □ ● Burns and Electrical Injuries □ ● Fractures Bandaging, immobilization, transferring etc.	8
IV.	**Environmental Emergencies and first aid:** ● Bites and Stings □ ● Poison Emergencies □ ● Heat-Related Emergencies □ ● Cold-Related Emergencies	7
V.	**Preparation of First Aid Kit**	1
	Total	**30 Hours**

B. Instruction Method and Pedagogy

The course is based on theory & practical learning. Teaching will be facilitated by reading material, discussion, microteaching, task-based learning, assignments, field visit and various interpersonal activities like group work, independent and collaborative research, presentations etc. Practical will be facilitated by demonstrations and preparing check-lists etc.

C. **Evaluation:** The students will be evaluated continuously in the form of internal as well as external examinations. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignments	1	8	8
2	Internal Test/ Model Exam	1	12	12
3	Attendance and Class Participation	Minimum 80% attendance		10
			Total	**30**

External Evaluation

The University Practical examination will be of 70 marks and will test the logic and critical thinking skills of the students by asking them to demonstrate or putting scenario based application of knowledge l avoid, as far as possible, grammatical errors and will focus on applications.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical cum Viva-Voce	01	70	70
			Total	70

D. **Learning Outcomes:** At the end of the course, learners will be able to:

- Demonstrate bandaging and immobilization of patient
- Demonstrate the skill for transferring injured person
- Demonstrate skills to clear airway & ventilate the patient
- Demonstrate skills to perform CPR

RD701.01: INTRODUCTION TO ANALYTICAL TECHNIQUES

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. **Outline of the Course**

Sr. No.	Title of Unit	No. of hrs.
1.	HPLC	08
2.	TGA-DSC	09
3.	DLS	05
4.	PCR	08

Total Hours (Theory): 30

B. **Syllabus Topics:**

Sr. No.	Title of Unit	Topics
1.	High Performance Liquid Chromatography (HPLC)	Introduction to HPLC, Basic Principle of HPLC, Instrumentation for HPLC, Types of Detector used in HPLC, Column efficiency in liquid chromatography
2.	TGA-DSC	General Discussion, Thermogravemetry Analysis, Instruments available for Thermogravimetric analysis, Detailed Discussion, Principle and Applications in various fields of Science and Engineering. Data Analysis
3.	DLS	Concept of Dynamic light scattering and basics of Particle size analyzer, Data Analysis and applications of DLS
4.	PCR	Introduction and principle of Polymerase Chain Reaction (PCR), Principle of PCR, Primer designing, Detailed methodology of PCR, Modifications of PCR, Applications of PCR

C. Instructional Methods and Pedagogy:

The topics will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Unit tests will be conducted regularly as a part of continuous evaluation and suggestions will be given to student in order to improve their performance.

D. Student Learning Outcomes / objectives:

- The Programme aims at providing students with the methodological concepts and tools needed to acquire top-level skills in the field of some selected instrumentation
- At the end students would gain experience in using these tools and analyzing the data.

E. Recommended Study Material:

❖ **Text books/Reference books**

1. Instrumental methods of analysis by Williard Merritt Dean Settle, 7th Ed. CBS publishers and distributors Pvt. Ltd.,
2. Instrumental methods of analysis by Williard Merritt Dean Settle, 7 th Ed. CBS publishers and distributors Pvt. Ltd.,
3. Principles of Gene Manipulation and Genomics by Sandy B. Primrose, Richard Twyman 7 th Ed. Wiley-Blackwell
4. Molecular Cloning: A Laboratory Manual by Joseph Sambrook, David William Russell 3 rd Ed.CSHL Press
5. Principles and Techniques of Biochemistry and Molecular Biology by Wilson and Walker 7 th Ed. Cambridge University Press
6. Principles of Instrumental Analysis by Douglas A. Skoog, F.James Holler and Timothy A.Nieman. Publisher: Saunders College Publishing.

RD 702.01: INTRODUCTION TO NANOSCIENCE & TECHNOLOGY

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of the Course:

Sr No.	Title of Unit	Minimum No. of Hrs
1.	Nanotechnology – development history & Implications of nanotechnology	6
2.	Overview on characterization and synthesis of nanostructure materials	10
3.	Overview of nanostructures & Nano devices	10
4.	New fields of nanotechnology	4

Total Hours (Theory): 30

B. Syllabus Topics:

1. Nanotechnology – development history & Implications of nanotechnology
   Concept of nanoscience and nanotechnology, Nanotechnology in the history and in nature. Impact of the nanotechnology in the society: Ethical, social, economic and environmental implications. the nanoscale. Size dependent physical and chemical properties. Surface effects. Importance of the surface at nanoscale. The surface/volume ratio. Size dependent properties
2. Overview on synthesis and characterization of nanostructure materials.
   Physical, chemical and biological methods of synthesis of nanostructures, Electron microscopy, scanning probe microscopy, non-imaging techniques
3. Overview of nanostructures & Nano devices
   zero dimensional, one dimensional and two dimensional nanostructures , Electronic devices, magnetic devices, photonic devices, mechanical devices, fluidics devices and biomedical devices
4. New fields of nanotechnology
   quantum computing, spintronics, nanomedicines, energy, etc.

C. Instructional Methods and Pedagogy:

The topics will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Students will be exposed to practical operations, and lab visit and experimental demonstrations of some of the equipment facilities for Nano fabrication & characterization available in UNI. Students will produce a technical report on the experiences.

D. Student Learning Outcomes / objectives:

Nanotechnology promises to be the technology of the future benefitting the humanity in a number of ways. This course is aimed at preparing students for further industrial or academic work in the field of nano-characterization techniques.

E. Recommended Study Material:

❖ **Text Books / Reference Books:**

1. Essentials of nanotechnology by Jeremy Ramsden [JR], 2009, Jeremy Ramsden & Ventus Publishing ApS,
2. Introduction to Nanoscience, S.M.Lindsay, Oxford ISBN 978-019-954421-9 (2010).
3. Guozhong Cao (2004). *Nanostructures and Nanomaterials: Synthesis, Properties & Applications***,** 448 pages, Imperial College Press, ISBN-10: 1860944159.
4. NANO: The Essentials Understanding Nanoscience and Nanotechnology, T. Pradeep, Tata McGraw-Hill Publishing Co. Ltd., 2007.
5. Nanoscience and Nanotechnology, B K Parthsarathy, ISHA Books, New Delhi, 2007.
6. Nanotechnology: Principles and practices, Sulabha K Kulkarni, Capital publishing company, 2007.

MB650: CREATIVE LEADERSHIP

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of the Course:

Module No.	Title/Topic	Classroom Contact Sessions
1	**Introduction to Leadership** • What are Leadership Skills? • Ways of Conceptualizing Leadership • Definition and Components • A Born Leader • Traits of Successful Leader • Why Leadership? ○ Managerial Roles • Importance of Leadership **Leading Vs Managing** • Roles and Relationships • Developing Personality for Effective Leading Roles • Authority Vs. Responsibility • Leading the Team • Leadership–Styles, Models and Philosophy	**05**
2	**Leadership Approach** • Trait Approach • Skills Approach • Style Approach • Situational Approach • Psychodynamic Approach **Leadership Theories** • Contingency Theory • Path-Goal Theory • Leader-Member Exchange Theory	**05**
3	**Leadership Processes** • Transactional Leadership • Transformational Leadership • Authentic Leadership • Team Leadership	**05**

Module No.	Title/Topic	Classroom Contact Sessions
	• Integrative Leadership • Liquid Leadership	
4	**Women and Leadership** • Gender and Leadership Styles • Gender and Leadership Effectiveness • Glass Ceiling Turned Labyrinth • Strengths of Women Leadership • Criticism and Application	**05**
5	**Culture and Leadership** • Dimensions of Culture • Clusters of World Cultures • Leadership Behavior and Culture Clusters • Universally Desirable and Undesirable Leadership Attributes • Criticism and Application • Leadership for High Performing Organisations • General Principles for Creative Culture • Nurturing Personal Creativity	**05**
6	**Contemporary Issues in Leadership** • Power and Politics in Leadership • Ethics in Leadership • Cases in Leadership	**05**
	Total	**30**

B. Pedagogy

The course will emphasise self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will be as follows:

- Classroom Contact Sessions … About 20 Sessions
- Case Discussions … About 03 Sessions
- Presentation … About 03 Sessions
- Management Exercise/ Stimulations/Game … About 02 Sessions
- Feedback … About 02 Sessions

The exact division among the above components will be announced by the instructor at the beginning of the semester as a part of detailed session-wise schedule.

C. Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sr. No.	Component	Number	Marks per incidence	Total Marks	Percentage of total internal evaluation
1	Quizzes	3	10	30	10
2	Case Analysis and Presentation	2	45	90	30
3	Assignment / Project work	1	60	60	20
4	Internal Tests	2	45	90	30
5	Attendance and Class Participation			30	10
			Total	**300**	**100**

The total marks will be divided by 10 and declared as Institute-level evaluation marks for the course. The Institute-level evaluation will constitute 30% of the total marks for the course.

D. External Evaluation

The University examination will be based on oral presentation, review of students' reports and a viva-voce and will carry 70% marks for the course evaluation.

E. Learning Outcomes

At the end of the course, the student should have developed:

- Appreciation for types, traits, approaches and leadership models/theories.
- Motivation for leadership roles and responsibilities.
- Qualities of creative leadership skills.

F. Reference Material

- **Text-book:**
  1. Leadership – Theory and Practices , Peter G. Northouse, Sage Publications India Pvt. Ltd., Latest Edition
- **Reference Books:**
  1. Liquid Leadership by Brad Szollose, Prolibris Publishing Media, Latest Edition
  2. Effective Leadership by Lussier/ Achua , Cengage Learning Publications, Latest Edition
  3. Integrative Leadership by Hatala & Hatala, Pearson Power Publication, Latest Edition
  4. Cases in Leadership by Rowe and Guerrero, Sage Publications India Pvt. Ltd., Latest Edition

- **Journals / Magazines / Newspapers:**
    1. HBR Issues on Building Leadership Skills
    2. International Journal of Innovation, Creativity and Change
    3. Economic Times
    4. Business Standard

PH891: COMMUNITY PHARMACY OWNERSHIP

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of course:

Sr. No.	Title of Unit	No. of Contact Hours
1	Introduction to Community pharmacy	10
2	Community Pharmacy Management	10
3	Pharmacy Business Plan	10
	Total	30

B. Detailed Syllabus:

Sr no	Title of Unit	Topics
1	Introduction to Community pharmacy	Roles & Responsibility, relationship with other health care providers, Prescribed medication order interpretation including OTC medicines, Safe use of medical devices.
2	Community Pharmacy Management	Role, process and scope of Community Pharmacy Management, modern technologies, financial, material, staff management and Drug store management, Code of ethics for Pharmacy.
3	Pharmacy Business Plan	Creating a Successful Pharmacy Business Plan, Business Owner Roles, Responsibilities, and Management Styles, Legal, Financial and Accounting Advice for the Beginning Owner, Marketing of Pharmacy Practice.

C. Instructional Methods and Pedagogy:

The content of the syllabus would be transmitted through different pedagogy tools like interactive class room sessions using classical chalk - board teaching to Power point presentations. Class room teaching would also be supplemented with group discussions, seminars, assignments and case studies.

D. Student Learning Outcomes / Objectives

At the end of the course, the student will be able to understand

- The concept of Community Pharmacy Ownership and its scope
- Students understand the role of the pharmacist in community and development
- Able to learn skill require to set Pharmacy store & its management

E. Recommended Study Material

❖ **Text / Reference Books:**

1. Mohd. Aquil, Practice of Hospital, clinical & Community Pharmacy Elsevier
2. Paul Rutter, Community Pharmacy E-Book, Symptoms, Diagnosis and Treatment, 3rd Edition
3. A Textbook of Clinical Pharmacy Practice: G. Parthasarathi, Karin Nyfort-Hansen and Milap Nahata, Universities Press.
4. Research articles as per the assignment

ME781.01: OCCUPATIONAL HEALTH & SAFETY

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	NA	2	2	2
Marks	NA	100	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to occupational health and safety	02
2	Occupational safety and management standards, and regulations for health, safety and environment	04
3	Identifying safety and health hazards, and risk analysis	04
4	Occupational physiology and psychosocial factors, and work organization	04
5	Ergonomic workplace design and musculoskeletal diseases	04
6	Control of workplace hazards	04
7	E-waste management	04
8	Work practices in industries and global strategy on occupational safety and health	04

Total Hours (Theory): 30

B. Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1	**Introduction to occupational health and safety**	**02 Hours**	**9%**
	Definition and history of occupational health & safety, workplace hazards.		
2	**Occupational safety & management standards and regulations for health, safety and environment**	**04 Hours**	**13%**

2.1	Factories act and rules; Workmen compensation act. Indian explosive act - Gas cylinder rules - SMPV Act - Indian petroleum act and rules.		
2.2	Environmental pollution act. Manufacturing, storage and import of Hazardous Chemical rules 1989, Indian Electricity act and rules. Overview of OHSAS 18000 and ISO 14000 National legislation and public organizations.		
3	**Identifying safety and health hazards, and risk analysis**	**04 Hours**	**13%**
3.1	Hazard, risk issues and hazard assessment, Introduction to hazard, hazard monitoring-risk issue.		
3.2	Hazard assessment, procedure, methodology; safety audit, checklist analysis, what-if analysis, safety review, preliminary hazard analysis (PHA), hazard operability studies (HAZOP).		
3.3	Computer aided risk analysis, Fault tree analysis & Event tree analysis, Logic symbols, methodology, minimal cut set ranking - fire explosion and toxicity index (FETI), various indices – Hazard analysis (HAZAN).		
3.4	Failure Mode and Effect Analysis (FMEA), Basic concepts of software on risk analysis, CISCON, FETI, ALOHA.		
4	**Occupational physiology, Psychosocial factors and Work organization**	**04 Hours**	**13%**
4.1	Man as system component – allocation of functions – efficiency.		
4.2	Occupational work capacity aerobic and anaerobic work – evaluation of physiological requirements of jobs – parameters of measurements – categorization of job heaviness		
4.3	Work organization – stress – strain – fatigue – rest pauses – shift work – personal hygiene.		
5	**Ergonomic workplace design and Musculoskeletal diseases**	**04 Hours**	**13%**
5.1	Meaning of Ergonomic		
5.2	Meaning of Workplace Design.		
5.3	Musculoskeletal Diseases causes and prevention		

6	**Control of workplace hazards**	**04 Hours**	**13%**
6.1	Workplace hazards and risk control, Transport hazards and risk control, Musculoskeletal hazards and risk control, Work equipment hazards and risk control		
6.2	Electrical safety, Fire safety, Chemical and biological health hazards and risk control, Physical and psychological health hazards and risk control, Health and safety practical application		
7	**E-waste management**	**04 Hours**	**13%**
7.1	Waste characteristics, generation, collection, transport and disposal		
8	**Work practices in industries and global strategy on occupational safety and health**	**04 Hours**	**14%**
8.1	Work practices in industries in manufacturing industries		
8.2	Work practices in industries in service industries		

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- Tutorials related to course content will be given to students.
- In the lectures discipline and behavior will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

D. Students Learning Outcomes:

- Students will be able to make models for safety at work.
- Students will be able to select safety methods.
- Students will be able to understand how hazardous the process is at work.
- Students will be able to understand proneness of accidents.

E. Recommended Study Material:

❖ Text Books:

1. Grimaldi and Simonds , Safety Management, AITBS Publishers , New Delhi (2001)
2. Industrial safety and health, David L. Goetsch, Macmillan Publishing Company, 1993.
3. R.K.Jain and Sunil S.Rao , Industrial Safety, Health and Environment Management Systems, Khanna publishers , New Delhi (2006)
4. Salvendy, G. (2012). Handbook of human factors and ergonomics. John Wiley & Sons.

❖ Reference Books:

1. Arezes, P., Baptista, J. S., Barroso, M. P., Carneiro, P., Cordeiro, P., Costa, N., & Perestrelo, G. (Eds.). (2013). Occupational Safety and Hygiene. CRC Press.
2. Chaturvedi, P. (2005). Managing Safety Challanges Ahead. Concept Publishing Company.
3. Healey, B. J., & Walker, K. T. (2009). Introduction to occupational health in public health practice (Vol. 13). John Wiley & Sons.
4. Hester, R. E., & Harrison, R. M. (2009). Electronic waste management (Vol. 27). Royal Society of Chemistry.
5. Karwowski, W., Soares, M. M., & Stanton, N. A. (Eds.). (2011). Human Factors and Ergonomics in Consumer Product Design: Uses and Applications. CRC Press.
6. Khan, B. H. (Ed.). (1997). Web-based instruction. Educational Technology.
7. Roughton, J., & Crutchfield, N. (2011). Job hazard analysis: A guide for voluntary compliance and beyond. Butterworth-Heinemann.
8. Salvendy, G. (Ed.). (2001). Handbook of industrial engineering: technology and operations management. John Wiley & Sons.
9. Smedley, J., Dick, F., & Sadhra, S. (Eds.). (2013). Oxford handbook of occupational health. Oxford University Press.
10. Tillman, C. (2006). Principles of occupational health and hygiene: an introduction. Allen & Unwin.

❖ Web Material:

1. International Labour Organization (ILO) http://www.ilo.org/public/english/
2. Occupational Safety & Health Administration United States Department of Laborhttps://www.osha.gov/about.html

❖ Other Material:

1. International Journal of Labour Research http://www.ilo.org/actrav/info/pubs/WCMS_158769/lang--en/index.htm

2. International journal of occupational safety And ergonomics (http://archiwum .ciop.pl/757).
3. Journal of Safety and Health at Work (http://www.journals.elsevier.com/safety-and-health-at-work/)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICAL SCIENCE

PD261 : Astrophysics, Space and Cosmos-1 (ASC–1)

University Elective
Semester 3 Undergraduate Programme
Semester 1 Postgraduate Programme

Prerequisite:

Student having exposure of 10+2 level of Physics, and studying in bachelor and master program from any institute can join this course.

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Basic Astronomical Techniques	10
2.	The Frontier of Space	06
3.	The Life and Death of Stars	08
4.	The Universe	06

Total Hours (Theory): 0

Total Hours (Lab): 30

Total Hours: 30

B. Detailed Syllabus:

1.	**Basic Astronomical Techniques**	**10 Hrs**
	Description of Telescopes, Methods of observation, electromagnetic spectral window, resolution, sensitivity, noise, signal to noise ratio, background, aberrations, Telescopes at different wavelengths, Detectors at different wavelengths, Imaging techniques, spectroscopy, calibration, Atmospheric effects at different wavelengths, Astronomical data analysis, H-R diagram, sun and solar system, Sky Gazing.	6 (L) + 4 (P)
2.	**The Frontier of Space**	**6 Hrs**
	Outer reaches of earth's Atmosphere, Earth's Atmospheric Layers and Orbital Mechanism: Types of Orbits Launching of Satellites, Basics of Satellite subsystems, channel and link. Satellite Applications: Earth Observation, Scientific Study, Weather Forecast, Military Applications, GPS.	
3.	**The Life and Death of Stars**	**8 Hrs**
	Stars and Galaxies, The story of collapsing stars: Star formation and evolution, Interstellar Nebula, Red Giant, Planetary Nebulae, Planetary Dynamics, White Dwarfs, Red super Giant, Supernovae- Neutron star, Black hole, naked singularity. Gravitational Lensing, Accretion disks Around Compact Objects, Binary pulsar, Gravitational Waves.	
4.	**The Universe**	**6 Hrs**
	Universe in different scale: solar scale, Galactic scale, Cosmological scale, vastness of our universe, Expanding universe (Big bang, inflation), some interesting facts of our universe.	

C. Instructional Methods and Pedagogy

The topics will be discussed in interactive class room sessions using classical black-board teaching, ICT, hands-on-experiments and demonstration of experiments, whichever is relevant. Assignments, small projects and lab-exercise will be to the students. Student’s needs to submit solution/report/results of above mentioned work, which will be eventually used for the evaluation purpose. Occasionally seminar/viva presentation will also be used as one of the evaluation tools.

D. Student Learning Outcomes:

Students will get preliminary knowledge about Astrophysics, Space and Cosmology. This is a level one course, which provides basis for the second level course. After completion of this course, students can start taking small project in this or allied fields.

F. Recommended Study Materials

- **Reference Book & Text Book:**

1. The Story of Collapsing Stars: Black Holes, Naked Singularities, and the Cosmic Play of Quantum Gravity, Prof. Pankaj S. Joshi
2. An Introduction to Cosmology 3ed, J.V. Narlikar
3. An introduction to modern cosmology, Andrew R. Liddle
4. Astronomy: Astronomy for Beginners: The Magical Science of Stars, Galaxies, Planets, Black Holes, Wormholes and much, much more! (Astronomy, Astronomy Textbook, Astronomy for Beginners), Miles Clarke
5. Satellite Communication, Dennis Roddy, Mc-Graw Hill publication
6. Satellite Communication, Timothy Pratt, Charles Bostian, Jeremy Allnut, John Wiley and Sons publication

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

List of University Elective Courses
For Second Semester Postgraduate Programme
(Academic year 2019-20)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	EE782.01: Energy Auditing and Management	EE/CSPIT
2	CE771.01: Project Management	CE/CSPIT
3	IT771.01: Cyber Security & Laws	IT/CSPIT
4	CA842:Mobile Application Development	CMPICA
5	PT796.01: Fitness and Nutrition	ARIP
6	NR 752.01: Epidemiology & Community Heath	MTIN
7	OC733.01: Introduction to Polymer Science	PDPIAS
8	MB651: Software based Statistical Analysis	I2IM
9	MA772.01: Design of Experiments	PDPIAS
10	PH892: Intellectual Property Rights	RPCP
11	PD262: Astrophysics, Space and Cosmos-2 (ASC-2)	PDPIAS

EE782.01: ENERGY AUDITING & MANAGEMENT

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the Course:

Sr. No.	Title of Unit	Min. No. of Hrs
1	**Electrical Energy Conservation**	3
2	**Electrical Energy Management**	10
3	**Financial Management**	4
4	**Energy Management & Audit**	5
5	**Energy Efficient Technologies In Electrical Systems**	3
6	**Energy Storage Systems**	3
7	**Case Studies.**	2

Total hours (Theory): 30

B. Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1	**Electrical Energy Conservation**	**3 Hours**	**11.33%**
1.1	**Energy Scenario:** Introduction to energy science and energy technology, various forms of energy. Law of conservation of energy. Energy scenario of India, Introduction to global energy scenario. Carbon credit, Energy Sector Reforms, Energy Strategy for the Future, Energy Conservation Act 2001 and its features.		
1.2	**Measures for energy conservation:** Potential energy conservation opportunities in: HVAC System, Lighting systems, Motors and Transformers.		
2	**Electrical Energy Management**	**10 Hours**	**29.12%**
2.1	Concept of energy management, Design of Energy management programmes, energy cost, Energy planning, Energy staffing, Energy Organization, Energy Requirement, Energy Costing, Energy Budgeting, Energy Monitoring, Energy consciousness, Energy		

	Management Professionals, Environment pollution due to energy use. Need of energy planning, steps for energy planning, Energy management in industry, Energy management cell function and objective, Energy management cell roles and responsibilities, Role of energy manager, benchmarking, Social and economic cost benefits. Seven principals of energy management.		
2.2	**Electrical System Optimization** Electricity rate tariff, key to reduction in electrical energy Consumption, Methods to improve plant power factor, load management, conduction loss, switching loss, magnetic loss, harmonic Compensation, Motor control, Lighting energy saving.		
2.3	**Cogeneration** Definition, Need, Application, Advantages, Classification, Saving potentials		
3	**Financial Management**	**4 Hours**	**11.11%**
	Investment-need, Appraisal and criteria, Financial analysis techniques-Simple payback period, Return on investment, Net present value, Internal rate of return, Cash flows, Risk and sensitivity analysis; Role of ESCOs.		
4.	**Energy Management & Audit**	**5 Hours**	**12.89%**
	Introduction, Definition, Energy audit- needs, types and walkthrough energy audit. Energy audit at unit level, Industrial Audit approaches. Procedure for energy audit and equipments required. Comprehensive Energy audit Site testing Measurement & Analysis of Electrical System like Induction Motors. Transformers, Illumination system, Problems on Energy Management.		
5.	**Energy Efficient Technologies In Electrical Systems**	**3 Hours**	**11.11%**
	Load Management and Maximum demand control. Electrical distribution system. Maximum demand controllers, Automatic power factor controllers, Energy efficient motors, Soft starters with energy saver, Variable speed drives, Energy efficient transformers, Electronic ballast, Occupancy sensors, Energy efficient lighting controls.		
6.	**Energy Storage Systems**	**3 Hours**	**13.33%**
	Introduction, Demand for energy storage, Energy storage systems: heat storage- hot water, hot solids, phase change materials; Potential energy storage: spring, compressed gas. Pumped hydro: Flywheels. Rolling mills, Electrical and magnetic energy storage systems.		
7.	**Case Studies.**	**2 Hours**	**11.11%**

C. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a component of the overall evaluation.
- Minimum two internal exams will be conducted and will be considered as a part of overall evaluation.

- Assignments based on course content will be given to the students for each unit/topic and will be evaluated at regular interval and its weightage may be reflected in the overall evaluation.

D. Student Learning Outcomes:

After the completion of the course the students will be able to:

1. Analyze about energy scenario nationwide and worldwide
2. Decide about energy management in more effective way.
3. To propose the effective way for energy conservation
4. Carry out financial management.
5. Can design energy efficient technologies and provide alternative solutions for energy storage

E. Recommended Study Material:

❖ **Text Book:**

1. Amlan Chakrabarti, Energy engineering and management, PHI Learning Private Limited.
2. K. Nagabhusan Raju, Industrial Energy Conservation Techniques, Atlantic Publishers & Distributors (P) Ltd.

❖ **Reference Book:**

1. Renewable energy sources and conservation technology By- N.K.Bansal, Kleemann and Meliss
2. Non – conventional energy sources by G.D. Rai
3. Energy technology by S.Rao.
4. A guide to energy management by Barney L Capehart, William J Kennedy, Wayne C Turner.

❖ **Web Material:**

1. www.energymanagertraining.com
2. www.bee-india.gov.in

CE771.01: PROJECT MANAGEMENT

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Overview of Project Management	03
2	Project Management Concepts and Techniques	03
3	Project Cost Estimation	04
4	Project Planning and Scheduling	05
5	Project Monitoring and Control	05
6	Material Management in Project	04
7	Management of Special Projects and Project	06

Total hours (Theory): 30

B. Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Overview of Project Management**	**03 Hours**	**10 %**
	Introduction to Project Management Overview of Project Planning, Project Estimation, Project Scheduling, Organization and Team Structure, Risk Analysis and Management, Resource Allocation Project Management Process and Role of Project Manager		
2.	**Project Management Concepts and Techniques**	**03 Hours**	**10 %**
	Project Screening and Selection Techniques Structuring Concepts and Tools (WBS,ORS,LRC) Project Planning Tools (Bar Chart, LOB, CPM and PERT) Risk Analysis and Management		
3.	**Project Cost Estimation**	**04 Hours**	**13 %**
	Types of Estimates and Estimating Methods Project Budgeting		
4.	**Project Planning and Scheduling**	**05 Hours**	**17 %**
	Dynamic Project Planning and Scheduling		

	Project Scheduling with Resource Constraints		
5.	**Project Monitoring and Control**	**05 Hours**	**17 %**
	Monitoring Techniques and Time control system Project Cost Control and Time cost tradeoff		
6.	**Material Management in Project**	**04 Hours**	**13 %**
	Project Procurement and Material Management		
7.	**Management of Special Projects and Project Management Software Tools**	**06 Hours**	**20 %**
	Management of SE/NPD/R&D/Hi-Tech and Mega Projects Software tools for Project Management: MS Project, Primavera, Turbo Project, Riski Project.		

C. Instructional Method and Pedagogy:

- Lectures will be taken in class room with the use of multi-media presentations and black board – mix of both.
- Assignments based on above course content will be given to the students at the end of each chapter. Each assignment contains minimum 5 questions.
- Quizzes and Surprise tests will be conducted for testing the knowledge of students for particular topic.

D. Student Learning Outcomes:

Upon successful completion of this course, students will be able to understand project management process and different aspect of development process necessary for the management of the project which includes various activities, resources, quality, cost and system configuration etc.

E. Recommended Study Material:

❖ **Text Books:**

1. Project management: engineering, technology, and implementation by Shtub, Avraham, Jonathan F. Bard, and Shlomo Globerson, Prentice-Hall, Inc., 1994.
2. Project Management Handbook by Lock, Gower.
3. VNR Project Management Handbook by Cleland and King.
4. Management guide to PERT/CPM by Wiest and Levy, PHI.
5. Project Management: A Systemic Approach to Planning, Scheduling and Controlling by Horald Kerzner, CBS Publishers, 2002.
6. Project Scheduling and Monitoring in Practice by S. Choudhury,

7. Total Project Management: The Indian Context by P. K. Joy, Macmillan India Ltd.

❖ **Reference Books:**

1. Project Management for Business and Technology: Principles and Practice by John M Nicholas, Prentice Hall of India, 2002.
2. Project Management, by N. J. Smith (Ed), Blackwell Publishing, 2002.
3. Effective Project Management by Robert K. Wysocki, Robert Back Jr. and David B. Crane, John Wiley, 2002.
4. Project Management: A Managerial Approach, by Jack R Meredith and Samuel J Mantel, John Wiley, 4th Edition, 2000.

IT771.01: CYBER SECURITY & LAWS

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	02	02	02
Marks	-	100	100	

A. Outline of the Course:

Sr No.	Title of the unit	Minimum number of hours
1.	Computer and Cyber Security Basics	06
2.	Security Threats	09
3.	Provisions in Indian Laws in dealing with Cyber Crimes	07
4.	Case Studies	08

Total hours (Theory): 30

B. Detailed Syllabus:

Sr. No.	UNIT	Hours	Weightage (%)
1.	**Computer and Cyber Security Basics**	**06 hours**	**20 %**
	Introduction to Computers, Computer History, Software, Hardware, Classification, Computer Input-Output Devices, Windows, DOS Prompt Commands, Basic Computer Terminology, Internet, Networking, Computer Storage, Computer Ethics and Application Programs, Security : Security trends –Goal, Attacks, Services and Mechanism		
2.	**Security Threats**	**09 hours**	**30 %**
	Application security (Database, E-mail and Internet), Data Security Considerations-Backups, Archival Storage and Disposal of Data, Security Technology-Firewall and VPNs, Intrusion Detection, Access Control. Security Threats -Viruses, Worms, Trojan Horse, Bombs, Trapdoors, Spoofs, E-mail viruses, Macro viruses, Malicious Software, Network and Denial of Services Attack, Security Threats to E-Commerce- Electronic Payment System, e-Cash, Credit/Debit Cards. Digital Signature, public Key Cryptography.		

3.	**Provisions in Indian Laws in dealing with Cyber Crimes**	**07 hours**	**23 %**
	Security Policies, Why Policies should be developed, WWW policies, Email Security policies, Policy Review Process-Corporate policies-Sample Security Policies, Publishing and Notification Requirement of the Policies. Information Security Standards-ISO, IT Act, Copyright Act, Patent Law, Cyber Laws in India; IT Act 2000 Provisions, Intellectual Property Law.		
4.	**Case Studies**	**08 hours**	**27 %**
	Identity Management, Cyber Security and Terrorism: Case Studies, The DigiNotar case, Deutsche Telekom, The disruption at the IT service provider Tieto, Web Based Attacks by Symantec, Password Security		

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- **Assignments** based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of **5 Marks** as a part of internal theory evaluation.
- Surprise Tests/Quizzes/Seminar/**Case Study** will be conducted which carries **10 Marks** as a part of internal theory evaluation.

D. Student Learning Outcome:

Learning outcomes of the course are:

- Students will able to do classification of cybercrime, methods used to perform crime, apply cyber security, and know the detailing of Information Technology Acts against offences.
- Students will understand and appreciate the legal and ethical environment impacting individuals as well as business organizations and have an understanding of the ethical implications of IT legal decisions.
- Students will have a fundamental knowledge of Information Technologies which affect organizational processes and decision-making.

E. Recommended Study Material:

❖ **Reference Books:**

1. Charles P. Pfleeger, Shari Lawerance Pfleeger, “Analysing Computer Security”, Pearson Education India.

2. V.K. Pachghare, “Cryptography and information Security”, PHI Learning Private Limited, Delhi India.
3. Dr. Surya Prakash Tripathi, Ritendra Goyal, Praveen kumar Shukla,”Introduction to Information Security and Cyber Law” Willey Dreamtech Press.
4. Schou, Shoemaker, “ Information Assurance for the Enterprise”, Tata McGraw Hill.
5. CHANDER, HARISH,” Cyber Laws And It Protection ” , PHI Learning Private Limited ,Delhi ,India

CA 842: MOBILE APPLICATION DEVELOPMENT

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the Course:

Week No	Practical	Description
1	Brief about mobile technologies and challenges in mobile application development and architectural overview of an Android platform.	Different mobile application development platform overview and Android architecture overview and basic components of Android application development overview.
2	Development of first Android based mobile application and overview of necessary components required for development.	Working of Android Studio IDE, Android project directory structure, Dalvik Virtual Machine Overview, Android Software development kit explanation, Virtual device creation and execution of first application on virtual as well as actual device
3	Fundamentals of User Interface designing.	XML based user interface designing using different available layouts like Relative Layout, Linear Layout, Table Layout etc.
4	User Interface Widgets-1	Hands-on demonstration of basic widgets like- TextView, EditText, Button, ToggleButton, RadioButton, RadioGroup, CheckBox, RatingBar, SeekBar etc**.**
5	User Interface Widgets-2	Hands-on demonstration of composite widgets like- ListView, Spinner and AutoCompleteTextView and customization of the composite controls.
6	Activity, Activity navigation and Intents.	Activity life cycle, Linking Activity using Intents:startActivity(), StartActivityForResult(). Calling built-in applications: ACTION_MAIN, ACTION_VIEW, ACTION_DIAL, ACTION_SEND
7	Android Resources, Styles and Themes.	Usage and implementation of different resources like drawable, string, color, dimes, raw and animation. Creating and Applying simple Style, Inheriting built-inStyle and User defined style, Using Styles as themes.
8	Dialogs & Menus.	Hands-on demonstration of different dialogs and menus available in Android.
9	Data Persistence Techniques.	User Preferences and Database management through SQLite

10	Broadcast Actions and Services implementation.	Service: life cycle, create and destroy service, Alarm Manager and SMS Manager. Standard Broadcast Actions.
11	PHP based web service implementation in Android	Creation and consumption of PHP based web service.
12	Simple Google Map incorporation with Android application.	Google Developer console usage, SHA-1 certificate creation and API-KEY creation and incorporation in Android application.

Total hours (Theory): 30

❖ **Text Books:**

1. Wei-Meng Lee: Beginning Android 4 Application Development, Wiley India Pvt Ltd.
2. Mark L. Murphy: The Busy Coder's Guide to Android Development

Reference Books:

1. Jonathan Simon: Head First Android Development, O'REILLY publication
2. Mark L Murphy: Beginning Android, Wiley India Pvt. Ltd.

Web References:

1. https://developer.android.com [Detail Android Development Guide]
2. https://www.youtube.com/watch?v=SUOWNXGRc6g&list=PL2F07DBCDCC01493A [200 android development tutorials]
3. www.androidhive.info/ [Advance application development with Android]

PT796.01: FITNESS AND NUTRITION

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the course:

S. No.	Title of the unit	Minimum number of hours
1.	Physical Education and Physical Fitness	10
2.	Nutrition and Health	10
3.	Sports and Life Skills Education	10

Total hours (Theory): 30

B. Detailed syllabus:

Sr. No.	UNIT	Hours
1	**PHYSICAL EDUCATION AND PHYSICAL FITNESS**	**10 hours**
1.1.	Concept of Physical Education, Meaning, Definition, Aims and Objectives of Physical Education, Need and Importance of Physical Education	
1.2	Physical Education and its Relevance in Inter Disciplinary Context	
1.3	Physical Fitness Components: Type of Fitness, Health Related Physical Fitness, Performance Related Physical Fitness	
2	**NUTRITION AND HEALTH**	**10 hours**
2.1	Concept of Food and Nutrition, Balanced Diet, Food Pyramid Index	
2.2	Macro and Micronutrients: Types, functions and classification system	
2.3	Carbohydrates : Types, RDA data, Glycemic index, Sources of Fats, saturated, unsaturated fats, recommended intake, importance of fat in diet, fats in health and disease	
2.4	Protein: Types EAA, function, assessing quality of proteins, selecting incomplete proteins, RDA sources.	
2.5	Vitamins And Minerals: Types, functions, sources, and minerals - calcium, Phosphorus	

	, iron, magnesium, sodium, potassium, and chloride. Trace elements - sources and functions.	
2.6	Determining Caloric Intake and Expenditure, Obesity, Causes and Preventing Measures – Role of Diet and Exercise, Importance of hydration in exercise	
3	**SPORTS AND LIFE SKILLS EDUCATION**	**10 hours**
3.1	Sports and Socialization	
3.2	Physical Activity and Sport – Emotional Adjustment and Wellbeing	
3.3	Substance Abuse among Youth – Preventive Measures and Remediation	
3.4	Yoga, Meditation and Relaxation	
3.5	Sports and Character Building	
3.6	Values in Sports	
3.7	Sports for World Peace and International Understanding	

C. Instructional Method And Pedagogy:

- Interactive classroom sessions using black-board and audio-visual aids.
- Using the available technology and resources for e-learning.
- Students will be encouraged towards self-learning and under direct interaction with course faculty.
- Students will be enabled for continuous evaluation.
- Case study, didactic mode of group discussions

D. Student Learning Outcomes:

Upon completion of the course, the student should be able to :

- Appraise the importance of exercise in maintenance of health and fitness.
- Objectively define health and fitness in realistic environment and inculcate habits of physical activity, nutrition and sports as a behavior change and overall health promotion

E. Recommended Study Material:

❖ **Textbooks:**

1. ACSM"s "Health Related Physical Fitness Assessment Manual Lippincott Williams and Walkins USA, 2005.
2. Siedentop.D,(1994) Introduction to Physical Education and Sports (2nd ed.), California: Mayfield Publishing Company.
3. Corbin.Charles Beetal. C.A., (2004) Concepts of Fitness and Welfare Boston McGraw Hill.

NR 752.01: EPIDEMIOLOGY & COMMUNITY HEALTH

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the Course:

Unit No.	Title of Unit	Prescribed Hours
1	**Introduction:** • Concept, scope, definition, trends, History and development of modern Epidemiology • Contribution of epidemiology • Implications	5
2	**Health Statistics:** • Morbidity & Mortality	2
3	**Epidemiological approaches:** • Study of disease causatives (Cause & Risk) • Health promotion • Levels of prevention	4
4	**Epidemiology of** • Communicable diseases • Non-communicable diseases	10
5	**Disaster:** • Disaster preparedness, • Disaster management	3
6	**Health Organizations:** • Voluntary health organizations • International health agencies –WHO, World health assembly, UNICEF, UNFPA, SIDA, US AID, DANIDA, DFID. AusAID etc	6
	Total	**30 Hours**

B. Instruction Method and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, microteaching, task-based learning, assignments, field visit and various interpersonal activities like group work, independent and collaborative research, presentations etc.

C. Evaluation:

The students will be evaluated continuously in the form of internal as well as external examinations. The evaluation (Theory) is schemed as 25 marks for internal evaluation and 75 marks for external evaluation in the form of University examination.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

<table>
<tr><th>Sl. No.</th><th>Component</th><th>Number</th><th>Marks per incidence</th><th>Total Marks</th></tr>
<tr><td>1</td><td>Assignments</td><td>1</td><td>8</td><td>8</td></tr>
<tr><td>2</td><td>Internal Test/ Model Exam</td><td>1</td><td>12</td><td>12</td></tr>
<tr><td>3</td><td>Attendance and Class Participation</td><td colspan="2">Minimum 80% attendance</td><td>10</td></tr>
<tr><td colspan="4">Total</td><td>30</td></tr>
</table>

External Evaluation

The University examination will be for 70 marks and will be based on practical based tests and a viva-voce.

D. Learning Outcomes: At the end of the course, learners will be able to:

- Understand the epidemiologic terminology,
- Understand the health statistic
- Understand the various methods of epidemiology
- Understand the role and functions of various health agencies
- Understand the epidemiological trends in communicable and non-communicable diseases.

OC733.01: INTRODUCTION TO POLYMER SCIENCE

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the course

Sr.No	Title of Unit	Approximate No of Hours
1	Basic concepts of Polymer Chemistry	5
2	Chemistry of Polymerization	8
3	Kinetics of polymerization	8
4	Molecular weight and size	9

Total Hours (Theory): 30

B. Detailed syllabus:

Sr. No	Title of Unit	Approximate No of Hours
1.	**Basic concepts of Polymer Chemistry**	**5**
	Introduction to polymers, How are polymers made, Classification of polymers.	
2.	**Chemistry of Polymerization**	**8**
	Chain polymerization, Step polymerization, Miscellaneous polymerization reactions, Polymerization techniques.	
3.	**Kinetics of polymerization**	**8**
	i). Free – Radical chain polymerization. ii). Cationic & anionic polymerization. iii). Polycondensation	
4.	**Molecular weight and size**	**9**
	Number – Average molecular weight. Viscosity – Average molecular weight. Polydispersity and molecular weight. Significance of polymer molecular weight. Size of polymer molecules	

C. Instrumental Methods and Pedagogy:

Topics will be taught in interactive class room sessions using black-board and if required power-point presentations will also be employed. Special interactive problem solving sessions will be conducted. Course materials will be provided from various sources of information. Students will be trained to measure molecular weight of polymers using appropriate instrument(s). Unit test will be taken regularly as a part of continuous evaluation and suggestions will be given to the students in order to do better in their performance.

D. Student Learning Outcomes/Objective:

- The programme aims at providing the basic concepts in polymer science.
- Ensuring that students acquire skills for further research in this area.

E. References:

1. A First Course in Polymer Chemistry by A. Strepikheyev , V.Derevitskaya and G.Slonimsky ; MIR Publishers, Moscow
2. Polymer Science by V.R.Gowariker , N.V.Viswanathan and Jayadev Sreedhar, New Age International Publishers.
3. Polymer Science and Technology of Plastics and Rubbers by Premamoy Ghosh, Tata McGraw-Hill Publishing Company Ltd. New Delhi.

MB651: SOFTWARE BASED STATISTICAL ANALYSIS

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**Introduction to Statistics** • Research and Innovation • Introduction to Statistics • Quantitative Techniques in Research	**06**
2	**Software # 1 - Use of Software in Research** • Introduction to the Software • Creating Variables • Data and its Types • Importing Data from MS-Excel • Transformation of Variables • Visual Benning • Determining Validity and Reliability of Scale using CFA - Cronbach's (alpha) : a coefficient of internal consistency • Estimation & Hypothesis Testing: • Parametric Tests ○ Z Test ○ t – Test ○ Cross Tabulation and Chi – square ○ One-Way & Two-Way Analysis of Variance (ANOVA) ○ Pearson’s Correlation Analysis ○ Regression Analysis ○ Simple & Multiple Linear Regression Analysis ○ Measures of Model Fit (R and R-square Statistics) • Non-Parametric Tests ○ Mann-Whitney U Test ○ Wilcoxon Signed Rank Test ○ Run test ○ Krushal-Wallis Test ○ Spearman Correlation Analysis	**12**
3	**Statistical Analysis Using Open Source software**	**10**

Module No.	Title/Topic	Classroom Contact Sessions
	• Introduction to Software • Programming Language Basics – including creating, sub-setting and analyzing • Managing your files and workspace • Controlling functions (procedures or commands) • Data Acquisition – Reading files • Data Transformations • Selecting variables and observations • Writing functions (macros) • Graphics	
4	**Article / Research Papers Reviews**	**02**
	Total	**30**

B. Pedagogy

The course will emphasise self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will be as follows:

- Classroom / Practical Contact Sessions … About 28 Sessions
- Research Paper Discussions / Feedback … About 02 Sessions

The exact division among the above components will be announced by the instructor at the beginning of the semester as a part of detailed session-wise schedule.

C. Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks	Percentage of total internal evaluation
1	Quizzes	3	10	30	10
2	Assignment / Project work	1	150	150	50
3	Internal Tests	2	45	90	30
4	Attendance and Class			30	10

	Participation			
		Total	**300**	**100**

The total marks will be divided by 10 and declared as Institute-level evaluation marks for the course. The Institute-level evaluation will constitute 30% of the total marks f or the course.

D. External Evaluation

The University examination will be for 70 marks and will be based on practical computer-based tests and a viva-voce.

E. Learning Outcomes

At the end of the course, the student should have developed:

- Skills related to use of statistical techniques for analysis using software
- Rational decision making skills for typical business / other decisions
- Inputs for reviewing articles / research papers especially related to use of statistical techniques and analysis based on software

F. Reference Material

❖ **Text-book**

1. Latest Manuals of Software

❖ **Reference Books**

1. David .M. Levine, Krehbiel, Berenson, P.K. Viswanathan, (Latest Edition), Business Statistics – A First Course, (Latest Edition), Pearson Education

MA 772.01: DESIGN OF EXPERIMENTS

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the course:

Sr No.	Title of the unit	Minimum number of hours
1.	Principles of Experimental Design	06
2.	Statistical Concepts	06
3.	Single Factor Experiments	08
4.	Factorial Experiments	10

Total hours (Theory): 30

B. Detailed Syllabus:

Sr. No.	UNIT	Hours
1	**Principles of Experimental Design** **06 Hours**	**20%**
1.1	Basic Terminologies: Types of Investigations and Experiments	
1.2	Confirmatory and Exploratory Experiments	
1.3	Modeling and selecting Response	
1.4	Minimizing Bias and Variability	
2	**Statistical Concepts** **06 Hours**	**20%**
2.1	Descriptive Statistics and Graphical Presentation	
2.2	Probability Distributions, Hypothesis Tests and Confidence Intervals	
2.3	Power and Sample size calculation	
2.4	Experiments for Two Treatments	
2.5	Linear Regression: Simple and Multiple	
3.	**Single Factor Experiments** **08 Hours**	**30%**
3.1	Completely Randomized Designs	
3.1	Concepts of Multiple comparison	

3.2	Pairwise Comparisons	
3.3	Comparisons with a Control	
3.4	General Contrast	
4.	**Factorial Experiments** **10Hours**	**30%**
4.1	Inference from Factorial Experiments	
4.2	Two-Level Factorial Experiments	
4.3	Definition and Estimation of Main Effects and Interactions	
4.4	Statistical Analysis	
4.5	Two-Level Fractional Factorial Experiments: Introduction	

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures/laboratory which carries a 5% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weighting of 5%.
- Two Quizzes (surprise test) will be conducted which carries 5% component of the overall evaluation.

D. Student Learning Outcomes:

- At the end of the course the students will be able to understand the basic concepts of Design of Experiment.
- Student will be able to apply concepts of these course in their study of specialization

E. Recommended Study Material:

❖ **Text Books:**

1. Tamhane, Ajit C. Statistical analysis of designed experiments: theory and applications. Vol. 609. John Wiley & Sons, 2009.
2. Hinkelmann, Klaus, and Oscar Kempthorne. Design and Analysis of Experiments, Introduction to Experimental Design. Vol. 1. John Wiley & Sons, 2008.
3. Lorenzen, Thomas, and Virgil Anderson, eds. *Design of experiments: a no-name approach*. CRC Press, 1993.

❖ **Reference Books:**

1. Cox, David Roxbee, and Nancy Reid. *The theory of the design of experiments*. CRC Press, 2000.

2. Goupy, Jacques L. *Methods for experimental design: principles and applications for physicists and chemists*. Vol. 12. Elsevier, 1993.

PH892: INTELLECTUAL PROPERTY RIGHTS

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the course:

Sr. No.	Title of Unit	No. of Contact Hours.
1	Intellectual Property Concepts	07
2	Practical aspect of patenting	20
3	IPR related treaties	03

Total Hours (Theory): 30

B. **Syllabus Topics:**

Sr. No.	Title of Unit	Topics
1	Intellectual Property Concepts	• Concept of property, conventional property vs. Intellectual Property • Basic aspect of the 8 different IPR mechanism Viz. Patents, Copyright, trademark, industrial design, layout design of integrated circuits, geographical indicators, plant varieties & trade secrets. • Benefits of IPRs to improve the quality of research work • Strategies for avoiding research duplications, infringements
2	Practical aspect of patenting	• Indian patent act and its recent amendment with respect to following aspect: Patentable and non-patentable inventions, Essential criteria for filing a patent, Filling a patent in India and abroad, Drafting of patent application • Patent Filling and Commercialization: Procedure for patent obtaining in India, National and International Patent Search, Patent Analysis, Patent Drafting and Filling Procedure in India, Patent Specification and Claims, How to right a Claim of Patent, Pre/Post Grant Issues in Patenting, Opposition of Patent Granting, Infringement Analysis, Ground of Defense, Intellectual Property Appellation Board (IPAB), International Filing: PCT System • Patent and Biodiversity Act

		• Introduction to World Intellectual Property Organization. (WIPO) • Commercialization of patent: Need for Commercialization of research and role of IPRs in research Commercialization. • Benefit/Disadvantages of patenting to the society • Latest Amendment/Emerging Issues in Patenting • Case Study
3	IPR related treaties	• Patent co-operative treaty • Budapest treaty

C. Instructional Methods and Pedagogy:

The course employs in interactive classroom session using chalk and talk teaching to power point presentations. It also includes presentation by students on a specific topic assigned to them by the faculty and case study discussion with various litigation, infringement and patent rejection cases. Unit test will be conducted regularly as a part of continuous evaluation and suggestion will be given to student in order to improve their performance.

D. Students Learning Outcomes/Objectives:

- At the end of the course, the student will be able to understand the fundamental concepts of Intellectual Property Rights which further will be helpful in understanding other advanced aspects of Patent and Trademark applications in various scientific research and innovation.
- At the end student would gain experience in filling and drafting procedure of Patent.

E. Recommended Study material

1. Intellectual Property Right basic Concept, by M. M. S. Khatri, Atlantic Publisher and Distributors Pvt. Ltd., New Delhi.
2. Epstein on Intellectual Property: 5th Edition by Michael M. Epstein, Wolters Kulwer India Pvt. Ltd. Gurgaon, India.
3. Intellectual Property Right and Human Development in India by Shabana Talwar, First Edition, Serials Publications, New Delhi, India.
4. IPR Handbook for Pharma Students and Researchers, Parikshit Bansal, Pharma Book Syndicate, Hyderabad, India.
5. Patents, N. R. Subbaram, Pharma Book Syndicate, Hyderabad.
6. Intellectual Property Right by Nikolaus Thumm, Springer-Verlag Publications, Germany.
7. Intellectual Property - Patents, Copyright, TradeMarks and Allied Rights by Cornish, Aplin and Llewelyn, Sweet and Maxwell – Thomson Publishers, New Delhi, India
8. The Enforcement of Intellectual Property Rights: A Case Book by Louis Tc. Harms, WIPO Publishing House, Geneva.
9. Intellectual Property: From Creation to Commercialization - A Practical Guide for Innovators & Researchers by John P., MC Manus, Oak Tree Press, Ireland.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICAL SCIENCE

PD262 : Astrophysics, Space and Cosmos-2 (ASC-2)

University Elective

Semester 4 Undergraduate Programme
Semester 2 Postgraduate Programme

Prerequisite:

Student having exposure of 10+2 level of Physics, and studying in bachelor and master program from any institute can join this course.

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Black holes and Shadows	10
2.	Gravitational Waves	06
3.	Sensors and Detectors for Satellite Imaging	08
4.	Cosmology	06

Total Hours (Theory): 0

Total Hours (Lab): 30

Total Hours: 30

B. Detailed Syllabus:

1.	**Black holes and Shadows**	**10 Hrs**
	Spacetime, Schwarzschild metric, Null Geodesics in Schwarzschild metric, Photon sphere, Einstein ring, Shadow of Black hole, shadow of Naked singularity, recent observation of black hole image (M87 Galaxy), experiments using imaging technique	8 (L) + 2 (P)
2.	**Gravitational Waves**	**6 Hrs**
	Introduction to Gravitational Waves; recent development for the detection of Gravitational waves; Overview of Gravitational Wave detectors (i.e. LIGO, KAGRA, etc.) and its scientific and technologic importance; Outline of Interferometers & its applications; Construction and working of Michelson Interferometer & its application potentiality in Advance- LIGO; Demonstration experiment of Michelson Interferometer.	4 (L) + 2 (P)
3.	**Sensors and Detectors for Satellite Imaging**	**8 Hrs**
	Introduction to sensors, Signal, Parameters, Types of Sensors: CCD and CMOS, CMOS sensors: Architecture Configuration and Working, CCD sensors: Architecture Configuration and Working.	
4.	**Cosmology**	**6 Hrs**
	Galaxies: General Description, The Milky-Way, M87 Galaxy, Rotation Curves of Spiral Galaxies, Galaxy Formation, Expanding Universe, Hubble's Law, Dark matter and Dark Energy.	

C. Instructional Methods and Pedagogy

The topics will be discussed in interactive class room sessions using classical black-board teaching, ICT, hands-on-experiments and demonstration of experiments, whichever is relevant. Assignments, small projects and lab-exercise will be to the students. Students needs to submit solution/report/results of above mentioned work, which will be eventually used for the evaluation purpose. Occasionally seminar/viva presentation will also be used as one of the evaluation tools.

D. Student Learning Outcomes:

After completion of Astrophysics, Space and Cosmology level one and two courses, students will explore themselves for taking projects, which may eventually fulfill their curiosity in these and allied fields. It may open up new avenues in their career.

E. Recommended Study Materials

❖ **Reference Book & Text Book:**

1. The Story of Collapsing Stars: Black Holes, Naked Singularities, and the Cosmic Play of Quantum Gravity, Prof. Pankaj S. Joshi
2. An Introduction to Cosmology 3ed, J.V. Narlikar
3. An introduction to modern cosmology, Andrew R. Liddle
4. Astronomy: Astronomy for Beginners: The Magical Science of Stars, Galaxies, Planets, Black Holes, Wormholes and much, much more! (Astronomy, Astronomy Textbook, Astronomy for Beginners), Miles Clarke
5. Satellite Communication, Dennis Roddy, Mc-Graw Hill publication
6. Satellite Communication, Timothy Pratt, Charles Bostian, Jeremy Allnut, John Wiley and Sons publication.
7. CMOS/CCD Sensors and Camera Systems, Second Edition, Gerald Holst, Terrence S Lomheim, SPIE publication.
8. Review article: Advanced LIGO, The LIGO Scientific Collaboration, Classical and Quantum Gravity, Volume 32, Number 7 (2015)